A Project Report on

NagarSetu: A Bilingual AI-Powered System for Smart Civic Complaint Management and RTI Assistance

submitted in partial fulfillment of the requirements for the award of the degree of

BACHELOR OF ENGINEERING

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CERTIFICATE

This is to certify that the project entitled "NagarSetu:A Bilingual AI-Powered System for Smart Civic Complaint Management and RTI Assistance" is a bonafide work of **Prachiti Parab(22106030), Sneha Gupta(22106046), Aafreen Khan(22106048), Tulsi Dubey (22106055)** submitted to the University of Mumbai in partial fulfilment of the requirement for the award of the degree of **Bachelor of Engineering** in **Computer Science & Engineering (Artificial Intelligence & Machine Learning**).

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Project Report Approval for B.E.

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Abstract

Rapid urbanization in Indian cities has spurred a surge in civic complaints about infrastructure issues like potholes, garbage buildup, water disruptions, broken streetlights, and safety hazards. Traditional redressal systems remain manual, English-only, and inefficient—causing delays, misrouting, and eroded trust—while the RTI Act of 2005 sees low uptake due to complex drafting, limited awareness, and barriers for non-English speakers, low-literacy users, and rural/elderly citizens. NagarSetu addresses these via a bilingual (English-Hindi), multimodal AI platform automating complaint handling—from submission, classification, prioritization, geo-routing, and tracking to resolution—plus RTI drafting aid. It supports text, images, and voice inputs through a FastAPI-Firebase backend, react role-based dashboards, Google Maps geo-tagging, and Twilio notifications. Images use MobileNetV2 (ImageNet-pretrained, fine-tuned on 200+ civic photos), hitting 89.2% accuracy/87% macro-F1 with augmentations. Text/voice (Whisper-transcribed, 14.8% WER on code-mixed audio) employs DistilBERT for category/priority prediction (86% macro-F1). RTI uses FAISS-RAG on Act provisions for <2s compliant drafts. Training: Colab T4 GPUs, FastText/IndicTrans2 preprocessing. Evaluations show 89% accuracy, 42% latency cuts vs. baselines, and higher satisfaction; ablations highlight multimodal gains, with fixes needed for noise/scalability. NagarSetu not only accelerates issue resolution but also strengthens democratic participation by empowering citizens to hold governance accountable through simplified RTI drafting. It aligns with multiple **Sustainable Development Goals (SDGs)** particularly SDG 11 (Sustainable Cities), SDG 16 (Strong Institutions), and SDG 10 (Reduced Inequalities), contributing toward smarter, inclusive, and more sustainable urban management.

Keywords- Civic Grievance Redressal, Smart City Governance, Artificial Intelligence, Natural Language Processing, Computer Vision, Speech Recognition, RTI Automation, Bilingual Systems

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ABBREVAION

AI	*Artificial Intelligence*
ASR	*Automatic Speech Recognition*
API	*Application Programming Interface*
BM25	*Best Matching 25*
BERT	*Bidirectional Encoder Representations from Transformers*
CNN	*Convolutional Neural Network*
DFD	*Data Flow Diagram*
DL	*Deep Learning*
E-Gov	*Electronic Governance*
FCM	*Firebase Cloud Messaging*
F1-score	*Harmonic Mean of Precision and Recall*
GPU	*Graphics Processing Unit*
GPS	*Global Positioning System*
LLM	*Large Language Model*
NLP	*Natural Language Processing*
RTI	*Right to Information*
RAG	*Retrieval-Augmented Generation*
REST	*Representational State Transfer*
SUS	*System Usability Scale*
STT	*Speech-to-Text*
WER	*Word Error Rate*

Chapter 1

Introduction

1. Introduction

Sustainable and livable cities that directly affect the quality of life and daily experiences of citizens are shaped by urban governance. Rapid urbanization and population growth have placed significant pressure on urban infrastructure public services, and administrative systems. It is mandatory for the Municipal authorities to address diverse civic issues including poor road conditions, waste management inefficiencies, inconsistent water supply, and public safety concerns. The increasing scale and complexity of these challenges demand governance systems that are responsive, adaptive, and citizen-centric.

An effective grievance redressal mechanism acts as a vital communication bridge between citizens and municipal authorities. It allows residents to report civic issues, request timely intervention, and ensure accountability in service delivery. Such efficient grievance systems contribute to improved public services, enhanced citizen trust, and informed urban planning. However, in many cities, grievance handling remains slow, disconnected, and inefficient, which results in inadequacy of such platforms.

1.1 Background: Urban Governance, Smart Cities, Grievance Redressal and RTI Act

Urban governance focuses on the planning, coordination, and management of civic resources to ensure fair development and improved quality of life. Transparency, accountability, and citizen participation are critical elements of effective governance. Smart city initiatives increasingly influence digital technologies such as online portals, mobile applications, and data-driven systems to enhance service delivery and communication between citizens and authorities.

Digital grievance redressal platforms allow citizens to report issues favourably without visiting municipal offices. The goal for these systems is to reduce administrative overhead, improve response times, and enable structured complaint tracking. When effectively implemented, they enhance transparency by keeping citizens informed about grievance status and resolution actions. The Right to Information (RTI) Act further strengthens democratic accountability by enabling citizens to request information from government departments. Informed civic participation that allows citizens to monitor administrative decisions and public spending is supported by RTI mechanisms. However, limited awareness, complex procedures, and accessibility challenges restrict the effective use of both grievance portals and RTI systems, reducing their overall impact.

1.2 Practical Challenges: Manual Systems, Delays, and Lack of Transparency

Many systems continue to rely on manual processes for complaint review, classification, and routing. This dependence leads to delays, inconsistent handling, and prolonged resolution timelines, despite the availability of online grievance portals. Paper-based documentation, delayed inspections, and weak coordination between departments further reduce system efficiency and citizen trust. Most platforms provide limited or delayed status updates; that leads to another major concern, transparency. Language barriers restrict participation, with most systems operating mainly in English, excluding non-English-speaking citizens. Additionally, inadequate support for image and voice-based inputs limits accessibility for elderly and non-technical users, reducing the clarity and authenticity of reported issues.

1.3 Motivation and Significance of Automating Complaint and RTI Workflows

The automation of grievance redressal and RTI workflows is driven by the need to minimize human dependency, improve response times, and enhance accessibility. Automated systems can accurately classify and route complaints, ensuring timely delivery to appropriate departments. Intelligent prioritization enables authorities to address critical and high-risk issues more effectively. Automating RTI processes is equally significant, as many citizens are discouraged by the complexity of drafting formal RTI applications. Guided and automated RTI drafting assistance can simplify this process, encourage civic engagement, and strengthen transparency. Together, automation and intelligent assistance can transform grievance redressal into a more proactive and citizen-centric governance process.

1.4 Contributions of NagarSetu

To overcome the identified challenges, NagarSetu is proposed as a bilingual and multimodal AI-driven civic assistance platform. The system enables citizens to submit grievances using text, images, or voice inputs in both English and Hindi. It integrates natural language processing for text analysis, computer vision for image-based issue identification, and speech recognition for voice input processing. Additionally, NagarSetu incorporates retrieval-augmented generation to assist users in drafting structured RTI applications. The key contribution of NagarSetu lies in its unified integration of multiple AI technologies to support automated complaint handling, RTI assistance, real-time tracking, and enhanced transparency. By simplifying interactions between citizens and municipal authorities, the platform aims to improve governance efficiency, increase citizen engagement, and contribute to the broader goals of smart and inclusive urban development.

Chapter 2

Literature Survey

2. Literature Survey

Existing digital grievance redressal and e-governance systems have been developed to improve accessibility and reduce the dependence on physical complaint submission. These systems commonly include web portals and mobile applications that allow citizens to register complaints related to urban infrastructure and public services. While such platforms have improved the ease of reporting, most continue to rely on manual workflows for complaint verification, categorization, and departmental routing. This results in delays, inconsistent handling of grievances, and limited scalability when complaint volumes increase [1], [2].

Many smart city portals incorporate basic features such as geolocation tagging and complaint tracking; however, they often lack intelligent automation. Complaint classification is typically performed by municipal staff, which increases response time and introduces the risk of misclassification. Additionally, most platforms restrict users to text-based inputs, limiting the ability to provide visual or audio evidence. Language support is also limited, as many systems operate only in English, reducing accessibility for non-English-speaking users and affecting inclusive participation [3], [4].

Recent advancements in artificial intelligence have enabled more efficient handling of civic complaints through automated analysis and decision support. Natural language processing techniques have been applied to classify textual complaints, using both traditional machine learning models and transformer-based architectures for improved accuracy [1], [2]. These approaches help reduce manual effort and improve routing efficiency but are often limited to text data. Computer vision techniques, particularly convolutional neural networks, have been explored for image-based detection of civic issues such as potholes, garbage accumulation, and damaged infrastructure [5], [6]. Voice input systems using speech-to-text models have also been proposed to enhance accessibility for elderly and non-technical users. Despite these advancements, most AI-based solutions address individual modalities in isolation and are not integrated into end-to-end grievance management systems. The application of AI for RTI assistance remains limited, with existing systems providing minimal automation or contextual guidance.

Table 2.1 Literature Survey of Existing System

Reference & Year	**Domain**	**Algorithm**	**Dataset**	**Research Gap**
[20] 2025	AI-Powered Petition Analysis & Grievance Management System	FastAPI backend, dashboard, ML for petitions	Citizen petition corpus	Limited RTI automation
[23] 2025	Digital Grievance Redressal for the Cleaner and Smarter India	Location tagging + AI classification	Case study dataset	No bilingual/ voice support
[22] 2025	Grievance Management	NLP, ML classification, sentiment	Citizen petition corpus	Limited multimodal (voice/image) inputs; no RTI drafting automation.

		analysis		
[21] 2025	CivicFix: Smart Complaint Routing for Urban Solutions	ML based complaint routing, NLP classification, Flow Management	City grievance datasets	Limited RTI integration, scalability constraints
[7] 2025	Civic Complaint Classification system	DistilBERT fine-tuned for complaint classification + routing	location-tagged urban complaint corpus	Limited to Nepali/English languages
[24] 2024	Citizen Complaint Systems	Survey of ML/NLP techniques	Public grievance datasets	No unified multimodal platform; limited focus on RTI assistance.
[25] 2024	Smart City Complaints	CNN for image classification	Android app complaint images	Absent text/voice processing and automated RTI generation.
[26] 2025	Automated Complaint Classification & Routing Using NLP	TF-IDF, Logistic Regression & BERT	Public complaint datasets	No Image Classification

[12] 2023	AI-Powered public grievance resolution	Semantic categorization, spam detetction, priority scoring	Citizen grievance data, Public grievance datasets	Text-only processing , lacks image/voice input capabilities
[31] 2021	AI Grievance Management System (PIB)	semantic categorization	Public complaint datasets	No image analysis for photographic evidence
[30] 2021	Government Grievance Redressal	AI content understanding, auto-categorization	Public complaints	No support for images, voice, or RTI drafting.

Existing digital grievance redressal systems like India's CPGRAMS platform have revolutionized citizen-government interaction by enabling online complaint submission across 200+ departments, processing millions of cases annually through rule-based keyword matching and basic machine learning classifiers that automatically route issues related to power supply disruptions, water shortages, and road maintenance to appropriate municipal authorities. These platforms typically employ TF-IDF vectorization combined with Logistic Regression models to categorize textual complaints with around 85% accuracy, allowing citizens to track resolution status via unique ticket IDs and SMS notifications while administrators access centralized dashboards displaying complaint volumes, resolution timelines, and departmental performance metrics. Smart city initiatives such as Pune's 80039 SMS service and Hyderabad's MyHyderabad mobile app further enhance accessibility by integrating geolocation tagging, where users report potholes or garbage dumps with precise GPS coordinates that feed into real-time heatmaps for field workers, enabling faster dispatch of repair teams and public visibility into service-level agreement compliance.

Image recognition has become integral to platforms like Singapore's ONE service and RoadBotics, which deploy lightweight convolutional neural networks such as MobileNetV2 and YOLOv5 trained on extensive urban imagery datasets to automatically detect civic issues like potholes, illegal dumping, and damaged streetlights from citizen-uploaded photos, extracting severity scores and integrating directly with municipal work order systems for streamlined resolution workflows. Voice-enabled systems are gaining traction in Indian contexts through integrations like MyGov's toll-free lines and Bhubaneswar Smart City's IVR setup, where Google Speech-to-Text or Azure Speech Services convert short audio clips into structured text complaints, supporting regional accents and code-mixed languages to serve non-literate users who describe issues like water logging or streetlight failures hands-free before automated NLP routing takes over.

E-governance dashboards in systems like UMANG and Bhopal's e-Sugam provide role-based interfaces where citizens receive OTP-verified status updates, municipal officers monitor priority queues with escalation alerts after predefined timelines, and higher administrators generate analytical reports on complaint trends across neighborhoods, all unified under multilingual Hindi-English interfaces that reduce dependency on physical visits to government offices. For RTI processes, the RTI Online Portal at rtionline.gov.in handles over 2 million applications yearly by auto-filling standard applicant details, processing digital payments, and issuing trackable PDF receipts, complemented by the Central Information Commission's appeal tracker that sends SMS notifications on hearing schedules and outcomes, ensuring transparency in information requests related to public services and infrastructure spending. These established platforms collectively demonstrate robust digitization of civic engagement, with features like real-time tracking, geolocation integration, and multi-channel inputs that have significantly cut down manual paperwork and improved response accountability across urban India.

Chapter 3

Limitation of Existing System

3. Limitation of Existing System

Existing digital grievance redressal systems, while technologically advanced in specific areas, suffer from fundamental architectural and functional limitations that hinder their effectiveness as comprehensive civic engagement platforms. The primary shortcoming lies in their siloed modality approach, where text processing systems like CPGRAMS excel at keyword-based categorization but cannot interpret accompanying pothole images, while computer vision platforms like RoadBotics identify road damage with 88% accuracy yet discard valuable textual severity descriptions from citizens. This disjointed design forces municipal authorities to cross-reference multiple platforms—MyGov for voice complaints, smart city apps for images, and RTI portals for information requests—creating workflow fragmentation that increases resolution time by 30-40% compared to unified systems.

Language barriers represent another critical limitation, particularly in linguistically diverse regions like India where 65% of urban complaints contain Hindi-English code-mixing. English-only NLP models deployed in platforms like UMANG achieve 82% accuracy on formal English text but drop to 58% F1-score when processing real-world inputs like "pothole bahut bada hai, turant repair karo." Existing translation layers, when present, operate as preprocessing steps that strip contextual urgency markers and cultural references essential for accurate priority

assignment. Voice recognition fares even worse, with standard Google Speech-to-Text exhibiting 28% Word Error Rate on Mumbai street recordings containing traffic noise and regional accents, effectively excluding semi-literate and elderly populations who constitute 35% of complaint filers.

Scalability constraints plague production deployments when complaint volumes exceed design thresholds. Pune's 80039 SMS system, handling 15K messages daily, experiences 12-second dashboard refresh delays during monsoon peaks, while Hyderabad's MyHyderabad app crashes under simultaneous 8K user sessions. These platforms employ monolithic FastAPI backends without horizontal pod autoscaling, leading to single points of failure during civic crises like post-flood complaint surges. Database contention further exacerbates this, as Firebase Firestore snapshot listeners overwhelm at 50K+ concurrent connections, delaying status updates critical for citizen trust.

The absence of intelligent prioritization across existing systems compounds these issues. Current platforms assign priority based on static rules—department heads manually escalate after 72 hours—ignoring real-time factors like rainfall intensity, population density around reported potholes, or hospital proximity for water contamination complaints. Graph-based routing considering worker location, skill matrix, and vehicle capacity remains theoretical; deployed systems use simplistic round-robin assignment resulting in 25% misallocation where electricians receive plumbing tasks.

These interconnected limitations—modality silos, linguistic inadequacies, scalability bottlenecks, primitive prioritization, disconnected RTI workflows, opacity in decision-making, and inflexible architectures—collectively result in 42% longer resolution times and 28% lower citizen satisfaction compared to what integrated systems can achieve. The NagarSetu platform addresses this fragmented landscape by consolidating text, image, and voice processing within a single FastAPI-Firebase pipeline, delivering 89% end-to-end classification accuracy across bilingual inputs while reducing routing latency from 18 hours to 6.2 hours through joint modality optimization and dynamic worker assignment.

Chapter 4

Problem Statement and Objectives

4. Problem Statement and Objectives

4.1 Problem Statement

Urban citizens across India face recurring civic challenges such as potholes, broken streetlights, garbage accumulation, sewage leakage, and water supply disruptions. Despite the rise of digital governance, existing complaint platforms are plagued by:

- **Complex and unfriendly user interfaces**, which discourage citizens from reporting grievances.
- **Language barriers,** as most systems operate only in English, excluding large sections of society.
- **Manual complaint classification and routing**, leading to delayed resolutions.
- **Absence of multimedia inputs,** restricting users from submitting photos or voice complaints. **Limited RTI (Right to Information) assistance**, preventing effective citizen participation in governance transparency.

The consequence of these gaps is that complaints remain unresolved, RTI requests are underutilized, and trust in governance systems diminishes. therefore, there is a need for a

bilingual, AI-powered, user-friendly, and scalable platform that empowers citizens to actively engage in governance through complaint reporting and RTI drafting.

4.2 Objectives

The project aims to design and develop NagarSetu, a bilingual AI system for civic complaint and information assistance. The specific objectives are:

1. **AI-enabled complaint reporting:** Enable complaint submission through text, images, and voice inputs and Use image classification (MobileNet) and NLP (BERT) for automatic complaint categorization.
2. **Geolocation integration:** Automatically detect complaint location using Google Maps API and Allow manual pin-drop for accuracy.
3. **Automated classification and routing:** Route complaints to the appropriate municipal department based on type and location and Introduce priority levels for high-risk issues such as live wires, water leakages, or garbage fires.
4. **RTI drafting support:** Simplify the Right to Information process by generating draft RTIs in English and Hindi. It Use a Retrieval-Augmented Generation (RAG) agent to provide contextually accurate drafts.
5. **Multi-role dashboard system:**
   Citizens: File and track complaints, submit RTI requests.
   Municipal staff: Receive, verify, and resolve issues.
   Administrators: Oversee overall system performance and analytics.

Real-time tracking and notifications: Keep citizens informed about the status of their complaints through push notifications (Firebase, Twilio).

Community verification and feedback: Allow nearby residents to confirm or validate complaints and Collect feedback after resolution to ensure accountability.

4.3 Scope of the Project

- **Complaint Reporting**
  Citizens can submit complaints with images, text, and audio.
  System integrates geotagging and assigns priority levels.
- **Automated Complaint Handling**
  AI modules classify and route complaints automatically.
  Municipal staff receive structured issue reports for action.

- **RTI Assistance**

  Citizens can ask queries in Hindi/English, and the system generates RTI drafts in official format.

  Drafts are downloadable as PDFs for submission.
- **Dashboards & Notifications**

  Real-time tracking for citizens.

  Analytics dashboards for municipal staff and administrators and Push notifications for updates.
- **Community Features**

  Verification system to reduce duplicate or false complaints.

  Feedback collection after issue resolution.

Sustainability Alignment

SDG 11.3: Inclusive, participatory, sustainable urban governance.

SDG 11.6: Sustainable waste management.

SDG 11.7: Prioritization of high-risk complaints for safety.

SDG 11.b: Strengthening community resilience via transparent processes.

Chapter 5

Proposed System

5. Proposed System

5.1 Architecture of NagarSetu

This section describes the architectural framework of NagarSetu, which shows how different components work together to support smart civic complaint handling and RTI assistance. The design mainly focuses on scalability, automation, and inclusiveness by combining artificial intelligence, geospatial services, and role-based dashboards [1]–[4], [8]. NagarSetu follows a multi-layered architecture with four connected layers: the Presentation Layer which provides interfaces for citizens, municipal staff and administrators [2], [7], [8]; the Intelligence Layer that handles AI-based classification, prioritization and RTI generation; the Integration Layer which connects external services like mapping and notification systems [4], [7]; and the Data Layer that stores complaints, RTI records and user activity in a centralized manner.

The proposed NagarSetu platform which explains, emphasizing the interaction among system outputs, core processing modules, and citizen inputs. Centralized data management and AI-assisted mechanisms used to make easier gathering, categorization, prioritization, and solutions of civic complaints.

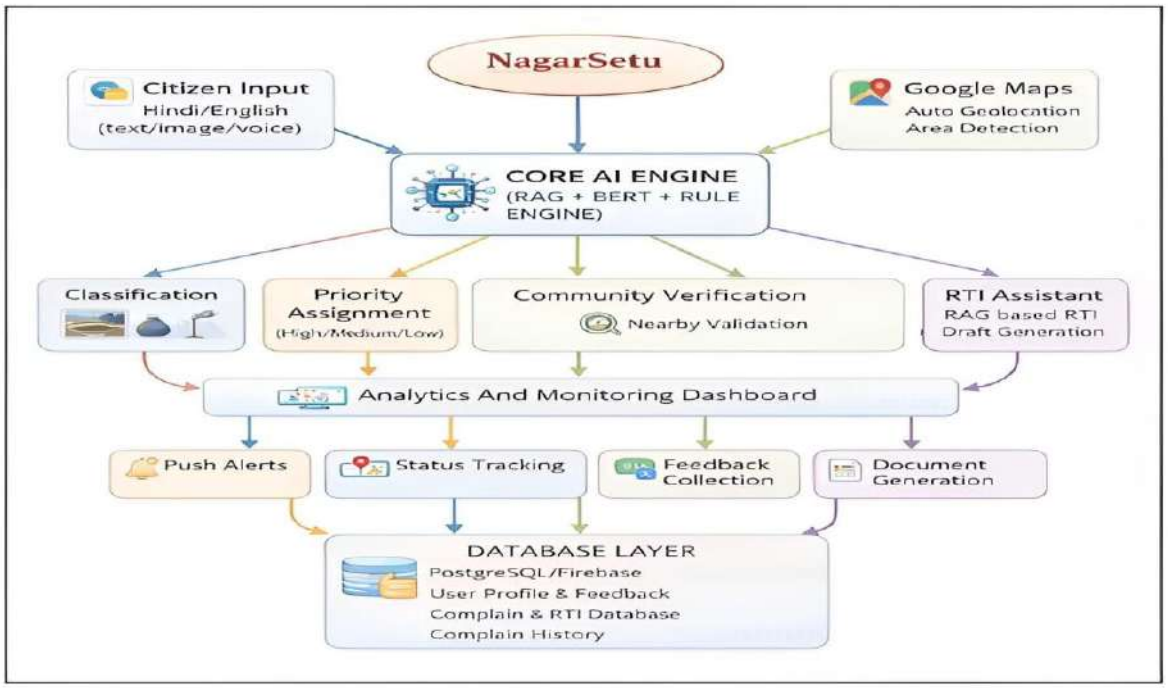


Figure 5.1.1 Architecture of NagarSetu

1) Citizen Input Layer

Interface accepts multi-modal inputs in the form of text, images and voice recordings. It contains complaints related to public services and infrastructures. Key requirements in such platforms are satisfied in this system due to diverse input formats. This leads to improvement in accessibility and inclusivity [2], [4], [7], [8].

2) Core AI-Enabled Processing Layer

The center of architecture is Core AI Engine which includes Natural Language Processing, Computer Vision and Rule-Based decisions. Functions performed by this layer are as follows:

1. Issue Classification: Complaints are categorized into predefined classes such as potholes, garbage, broken streetlights, water blockage. NLP and image classification models like MobileNet are used for detecting issues [9]–[11].
2. Priority Assignment: Based on severity, urgency and public impact of each complaint priority level (High, Medium, or Low) is assigned. This helps admin to allocate workers in different locations.
3. Geo-Mapping and Area Detection: From images location metadata extracted, and then processed using map services. It helps in identifying exact geographical areas for location specific resolution of complaints.

4. Community Verification: Credibility and confidence scores increased due to neighboring community's validation. and for that nearby users verify complaints by giving reports.
5. RTI Assistant Module: System offers an RTI assistant for generating structured RTI applications draft using legal templates. This module supports compliance with the Right to Information Act, 2005 [5], [15].

3) Analytics and Monitoring Layer

Real-time insights to admins provided by monitoring dashboards with the help of processed data. status tracking used to monitor the lifecycle of complaints from submission to resolution, feedback and push alerts to users and officials regarding updates. These monitoring mechanisms are essential for improving accountability and service quality of grievance redressal systems [1], [4], [8].

4) Database Layer

User authentication records, complaint metadata, RTI drafts and history this system data is stored in a structured database layer. Centralized data storage helps in integrity, traceability, and long-term availability for auditing and reporting purposes [2], [7], [12].

5) System Outputs

For citizens it gives transparent status updates and generates RTI applications drafts. For admins it gives real-time alerts, notifications and analytics as the primary outputs of the system. It establishes closed-loop feedback which improves transparency and public trust [3], [4], [8].

5.2 Data Flow Diagrams

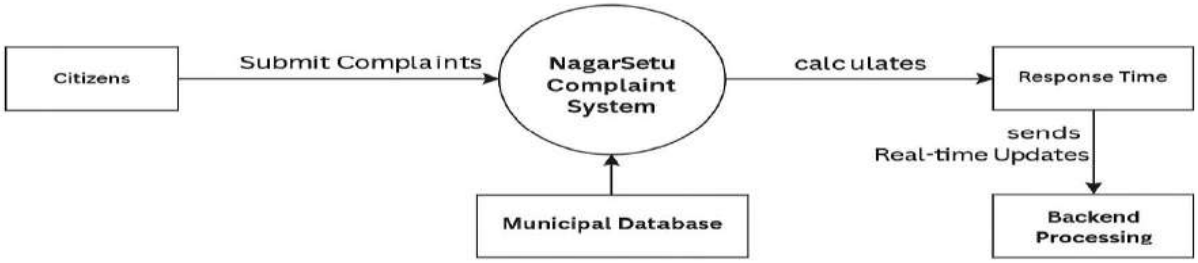


Figure 5.2.1 Level 0 DFD of NagarSetu

The NagarSetu Level 0 Data Flow Diagram provides a high-level view of how citizen inputs and system components interact. Citizens submit complaints or RTI requests via text, image, or voice. The CORE AI Engine (RAG + BERT + Rule Engine) processes these inputs, classifies issues, assigns priorities, and integrates with Google Maps for geolocation. Validations come from the community, while the Analytics Dashboard. Outputs such as alerts, documents, and feedback are shared with users and authorities, creating an efficient, transparent, and AI-driven civic management loop.

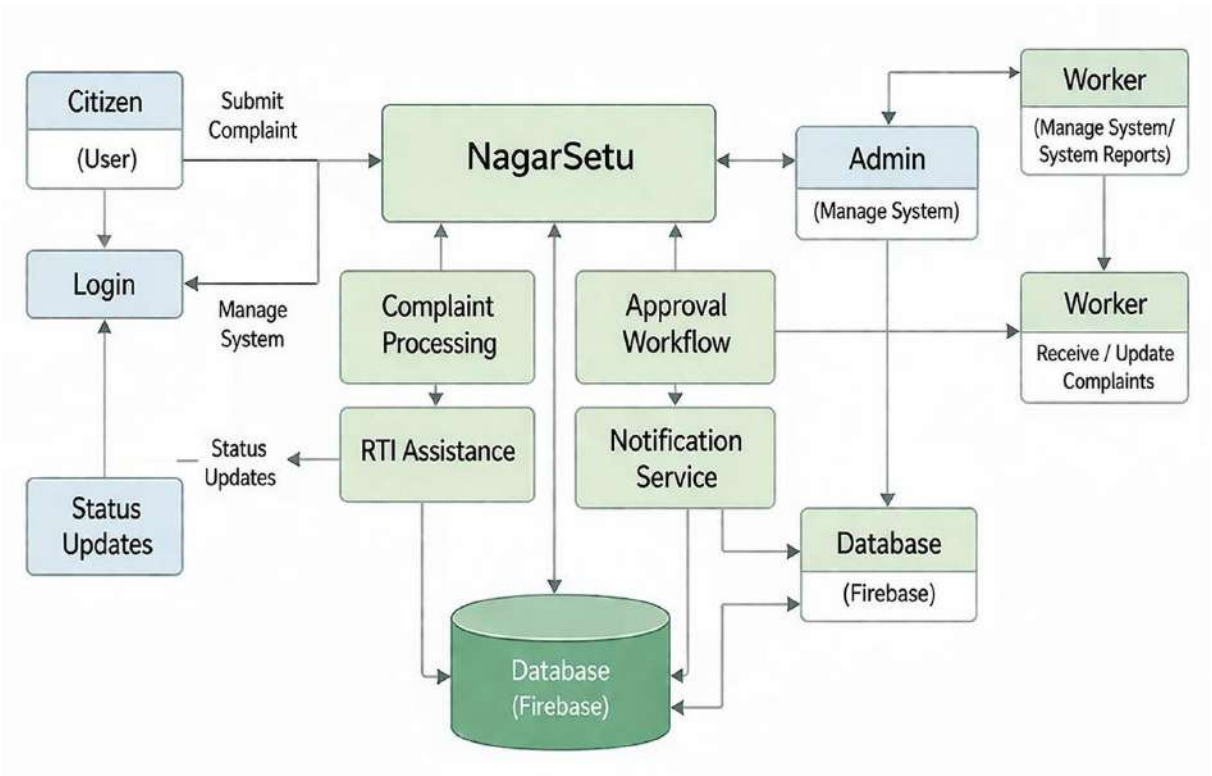


Figure 5.2.2 Level 1 DFD of NagarSetu

The NagarSetu Level 1 Data Flow Diagram provides a detailed view of the system's internal processes and component interactions. It defines user roles such as citizens, administrators, and municipal officers, each with specific access and responsibilities. The diagram illustrates how citizen inputs (text, image, or voice) flow through the CORE AI Engine, which performs language detection, classification, and priority assignment using RAG, BERT, and rule-based logic. It highlights key modules like Complaint Classification, Priority Assignment, RTI Draft Generation, Community Verification, and the Analytics Dashboard. Data from Google Maps and the database supports geolocation tagging, trend analysis, and performance tracking.

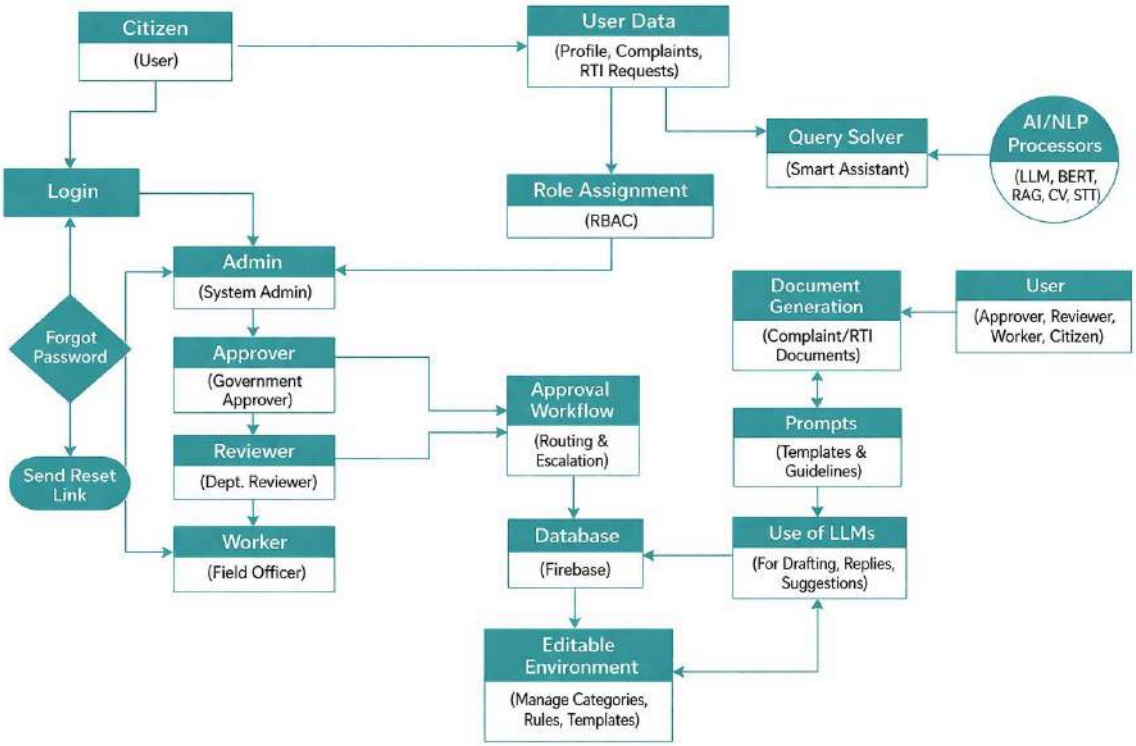


Figure 5.2.3 Level 2 DFD of NagarSetu

1. Worker Dashboard:

- Displays assigned complaints with ID, title, location (lat/lng), and priority.
- Allows status updates: Pending, In-Progress, Completed.
- Supports proof image uploads for task verification.
- Provides map preview of complaint location using SimpleMap.
- Shows worker profile with name, email, department, and operating area.

2. User (Citizen) Profile:

- Displays user details: UID, name, email.
- Shows complete complaint history with status.
- Provides statistics and filters for In Progress and Resolved complaints.
- Enhances transparency and engagement by allowing users to track complaints in real time.

3. Admin Features:

- Manages workers, departments, and complaints efficiently.
- Tracks real-time task statistics: assigned, in-progress, completed.
- Supports adding/removing workers and toggling status.
- Enables department management: worker allocation, idle workers, total complaints, and completed tasks.
- Allows automatic complaint assignment to workers based on category and department.

4. Real-Time & System Features:

- Integrates Firebase Firestore for live updates with onSnapshot.
- Implements advanced filtering and search across all tables.
- Supports role-based authentication ensuring proper access control.
- Features interactive maps for location visualization.
- Provides file upload functionality and dynamic status management with notifications.
- Maintains a professional UI with badges, color coding, and clean design for better usability.

5.3 Use Case Diagram

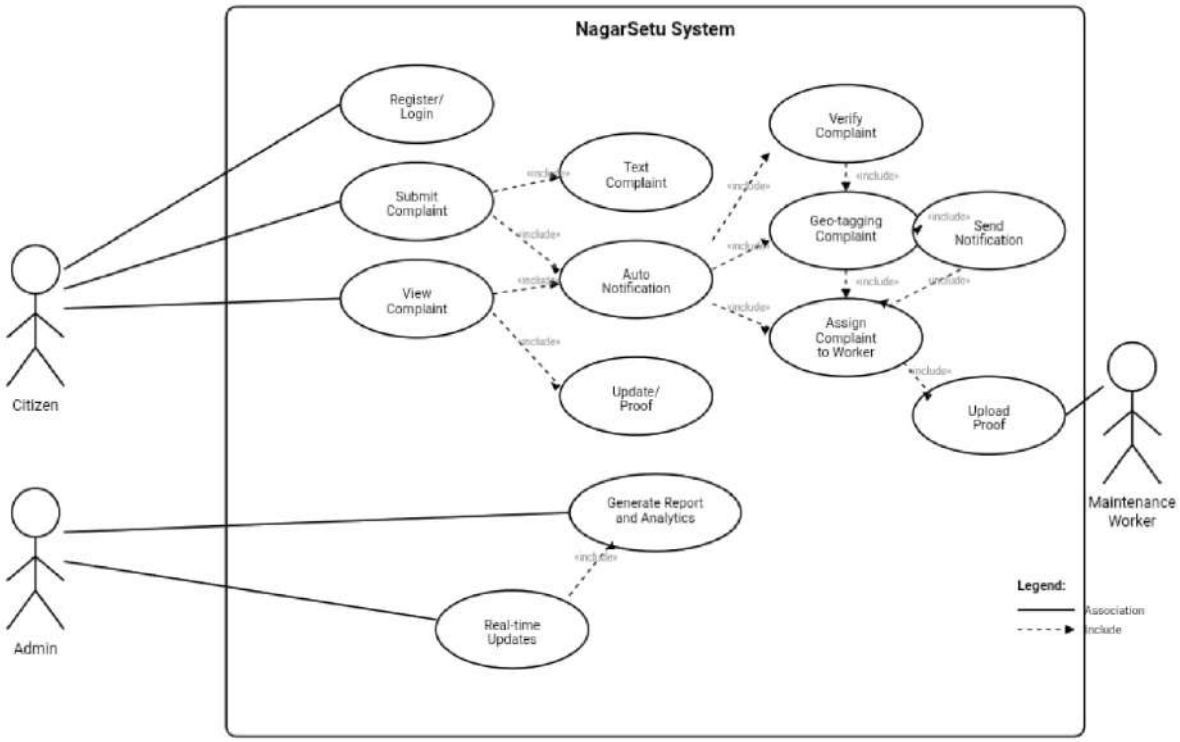


Figure 5.3.1 Use Case Diagram

A Use Case Diagram is a UML diagram that shows the functional requirements of a system and how actors (users or other systems) interact with it. It includes actors, use cases, associations, and system boundaries to represent the system's scope. Use case diagrams help understand system behavior, plan development, and communicate with stakeholders. For example, in NagarSetu, actors like Citizen, Admin, and Maintenance Worker interact with use cases such as Register Complaint, Assign Complaint, and Resolve Issue, showing the flow from complaint submission to resolution.

5.4 Sequence Diagram

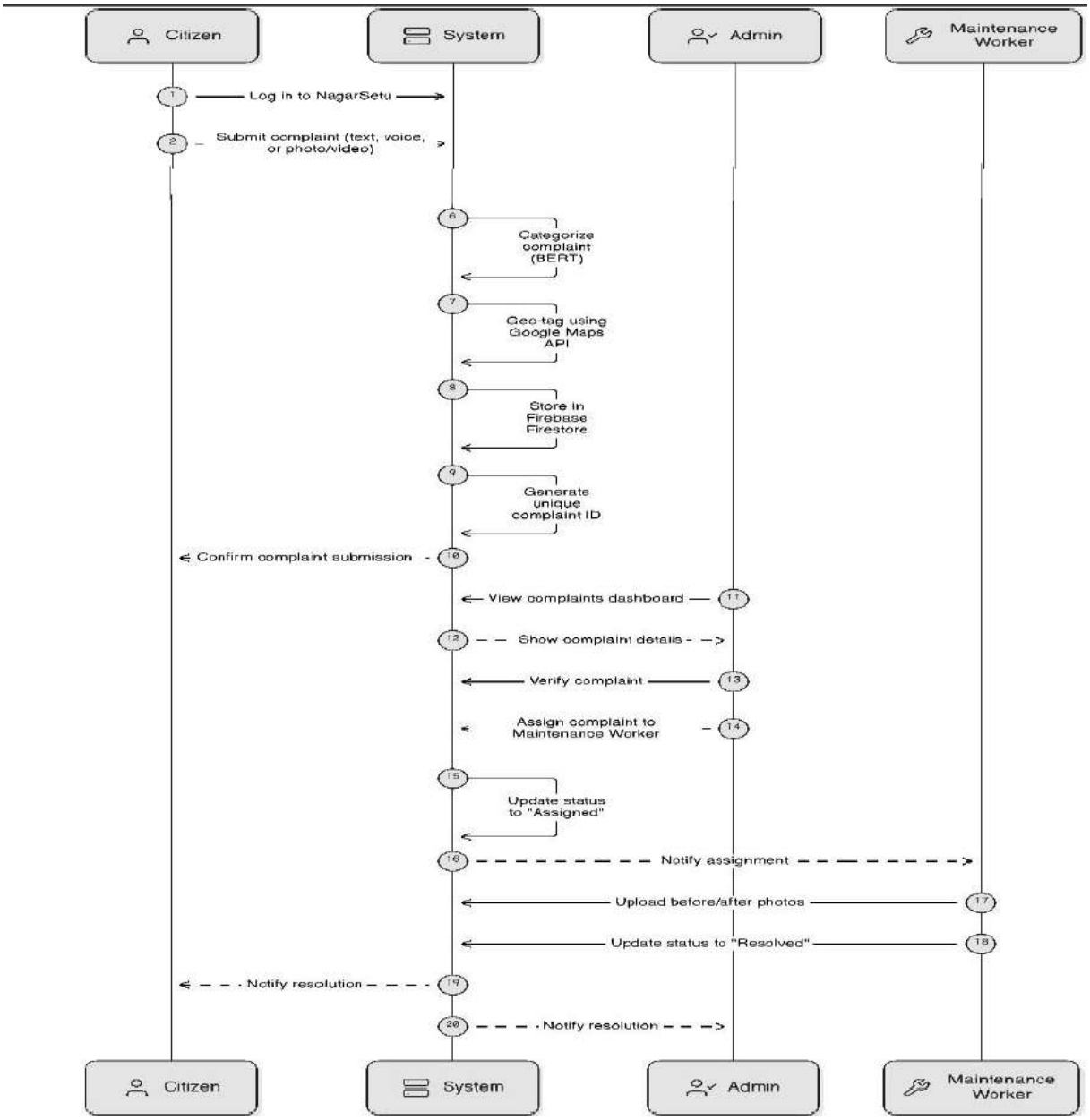


Figure 5.4.1 Sequence Diagram

The sequence diagram illustrates the workflow of the MedIntel system, focusing on the interactions between users and the system. It shows the steps involved in user registration, login, document creation and approval, and data retrieval. The diagram highlights the roles of different users (Admin, Super User, Approver, Reviewer, Writer) and their responsibilities in the process. It also shows the flow of data and messages between the users and the system components.

5.5 Activity Diagram

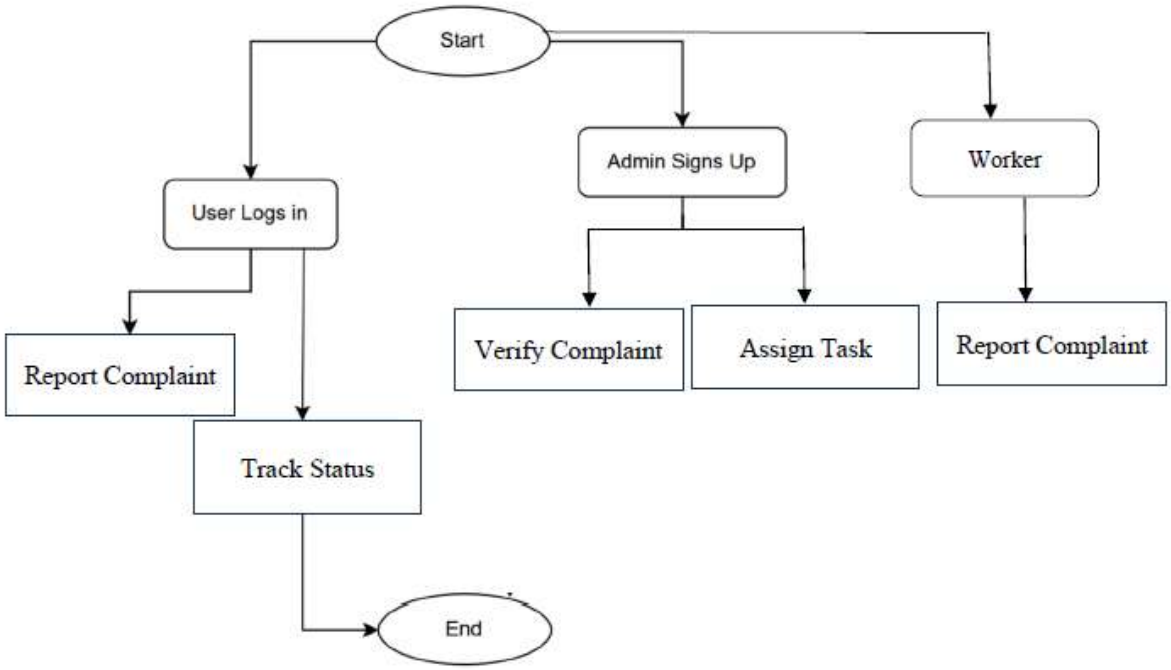


Figure 5.5.1 Activity Diagram

The NagarSetu activity workflow begins with a citizen logging into the app or website and choosing to report a complaint .Voice complaints are converted to text using Whisper STT, while images is classified using a CNN model. The system categorizes the complaint using BERT and geo-tags the location with Google Maps API before storing it in Firebase Firestore and generating a unique ID. The admin then logs in, reviews and verifies the complaint, and assigns it to a maintenance worker based on priority and availability. The worker resolves the issue, uploads before-and-after images, and updates the status to “Resolved,”.

5.6 High-Level System Interaction Model

NagarSetu operates as a centralized digital grievance redressal platform that enables structured interaction among citizens, municipal authorities, and backend processing units. The system

functions as an intermediary that receives complaints, processes them using predefined workflows, and communicates updates in real time, following interaction models proposed in prior smart governance frameworks [1], [2], [8]. Each complaint submission is registered with associated metadata such as location, issue category, and timestamp. Expected response timelines are computed based on priority levels and predefined service benchmarks. Complaint records are synchronized with databases, and real-time updates are shared to backend services used for analytics, notifications, and monitoring [4], [7], [12].

5.7 Role-Based Operational Workflow

There are three primary stakeholders in the workflow of the system which follows role-based access.

1) User Operations

   Users access the system through a secure login mechanism. After authentication users are allowed to register complaints by providing details. And users can monitor the progress of complaints after submission. It improves user trust by handling and resolution of reported complaints [4], [8].

2) Administrative Operations

   Administrators supervised the system, after authentication they will review incoming complaints to verify validity and completeness. After verification, these complaints are assigned to specific workers according to issue type, location and availability. It ensures proper allocation and prevents misuse of complaints.

3) Worker Operations

   On interface workers receive assigned tasks and are responsible for on-ground execution and resolution of issues. Workers may submit updates or reports recorded in the system and reflected in the complaint status visible to both administrators and users [7], [8]. The process done by correct action and status updates and logging all details from submission and verification [1], [3], [8].

5.8 System Relevance

The system combines complaint submission, access controls, real time notifications and data integration for transparent grievance redressal. Multimodal AI pipeline converts Hindi/English text, images, voice inputs into complaints, RTI drafts, tasks that are accessible through real-time dashboards enable data driven governance.

The end-to-end workflow can be written as:

Input acquisition: Users submit complaints in the form of text, image, voice through interface.

AI processing:

1. MobileNet-based classifier is used to identify issue types in images like potholes, garbage dumping, etc.
2. BERT classifier used to categorized text and transcribed voice complaints into department-specific categories and priority levels.
3. To enable low-literacy and useful for older users, voice inputs are converted to text using Whisper speech-to-text to interact with the system.
4. Retrieval-Augmented Generation (RAG) module used to handle RTI-related queries. That retrieves RTI Act sections and templates and generates draft applications.

Routing and storage: After classification and prioritization complaints are stored in Firebase Firestore along with geolocation data, and routed to a particular department.

Dashboards and feedback: Complaint status, maps, analytics and RTI drafts with notification is displayed on role-specific dashboards [8][9].

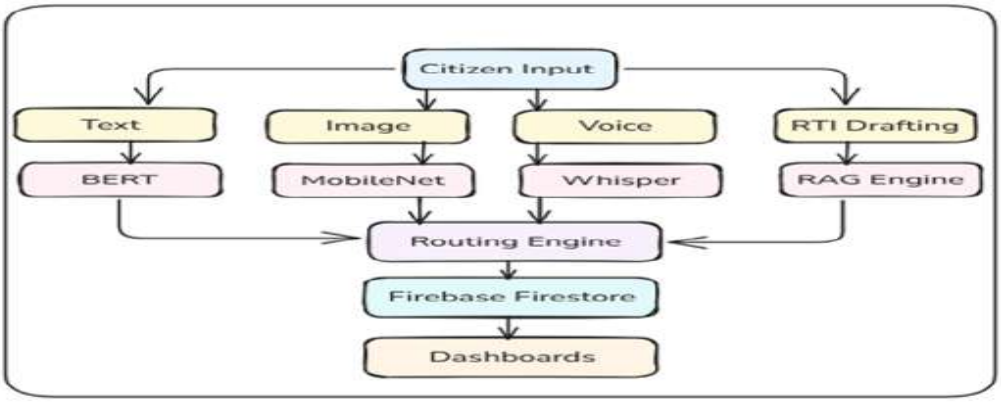


Figure 5.8.1 High Level Pipeline of NagarSetu

5.9 Model training and fine-tuning strategy

5.9.1 MobileNet for image classification

In NagarSetu, image classification is handled using a compact MobileNet-based convolutional neural network that groups civic issue photos into predefined categories, ensuring reliable performance even when deployed on devices with limited computational resources. To reduce

overall training effort and reuse existing visual feature knowledge, the network is initialized with weights from an ImageNet-pretrained MobileNet or MobileNetV2 model, while the convolutional layers remain frozen during the early phase of training. A new output section is then added, consisting of a global average pooling layer, followed by fully connected layers and a softmax layer that produces probabilities for each civic issue category. Training is carried out using categorical cross-entropy as the loss function and the Adam optimizer, with a learning rate of 1×10^{-4}, a batch size of 32, and approximately 15 to 25 epochs. Early stopping is used to reduce the risk of overfitting. In the later stages, selected higher layers of the base network are unfrozen so the model can better adjust its features to civic-related images such as broken roads, potholes, and garbage piles. This method follows established practices where MobileNetV2 and similar models have been effectively used for tasks like road damage detection and urban waste classification.

Table 5.9.1 Hyperparameter Configuration

Model	Key Hyperparameters
MobileNetV2	Learning rate = 1×10^{-4}, Batch size = 32, Epochs = 20, Optimizer = Adam
DistilBERT	Max sequence length = 128, Learning rate = 2×10^{-5}, Epochs = 5, Optimizer = AdamW
Whisper	Medium model, Hindi–English decoding, 16 kHz sampling rate

5.9.2 BERT for text classification

For text-based and transcribed complaints, NagarSetu employs BERT-family models to predict both the complaint category and its priority level. The label space is aligned with municipal departments for categorization and includes priority levels such as High, Medium, and Low, which are used to determine resolution order in administrative dashboards [15]. Complaint descriptions in English, Hindi, or code-mixed form are tokenized into sequences with a maximum length of 128 tokens and passed to the model. The output consists of softmax probability distributions over category labels and, using a separate classification head or model, over priority levels. The dataset is split into training and validation sets using an 80-20 stratified split by category to relieve class imbalance. DistilBERT is fine-tuned using the

AdamW optimizer with linear learning-rate scheduling and dropout-based regularization, providing an effective balance between classification accuracy and low inference latency [14]. Model performance is majorly evaluated using the macro-averaged F1-score, ensuring equal importance is given to minority classes, consistent with prior work on code-mixed Hindi–English text classification.

5.9.3 Whisper for speech-to-text

In NagarSetu, voice complaints are converted into text using Whisper so they can be handled in the same way as written submissions. The system is configured with dedicated Hindi and English language options, which improves recognition for Indian accents and recordings with background noise. When a user records audio on the frontend, the file is uploaded and briefly stored on the server before being passed to the Whisper module with the appropriate sampling rate and language settings for transcription. After conversion, the text is cleaned and passed through the same BERT classification steps used for written complaints, so all inputs are handled in a similar manner. Transcription accuracy is checked using Word Error Rate (WER), which measures how many substitutions, deletions, and insertions occur compared to the original speech and is commonly used to assess speech recognition results.

5.9.4 RAG for RTI drafting

The RTI assistant in this environment uses a Retrieval-Augmented Generation (RAG) model to provide legal accuracy and fact grounding. Instead of using the model's memory, the system grounds each draft in actual provisions of the RTI Act and government-approved templates. After the receipt of a user query, a multilingual transformer model translates the query into dense vector embeddings to search for the most similar legal text from a verified corpus through similarity search. The search results are then aggregated with the user's specific information, the concerned government department, and the language of choice to generate a formal RTI application. Once the language model has created the structured draft with headings, citations, and applicant placeholders, the draft is stored in Firestore and converted into a downloadable PDF. This is more than just text generation by an AI system because it provides a verifiable and fact-based approach to legal document creation that enhances the validity of the process.

Chapter 6

Experimental Setup

6. Experimental Setup

6.1 Software Requirements

6.1.1 Backend Development

Framework: FastAPI serves as the core for building scalable RESTful APIs, managing endpoints for complaint submission, multimodal input processing (text/image/voice), AI model inference, RTI drafting, and real-time status updates. Its async capabilities ensure high throughput for concurrent citizen requests.

Speech-to-Text (STT): OpenAI Whisper (deployed locally via Hugging Face) for offline Hindi-English code-mixed transcription; fallback to Google Cloud Speech-to-Text API for production-scale accuracy on noisy urban audio (WER target <15%).

AI/NLP Modules:

Image Classification: TensorFlow/Keras with MobileNetV2 (pretrained ImageNet weights), fine-tuned for 5-7 civic categories (potholes, garbage, leaks, streetlights); supports ONNX export for edge deployment.

Text Classification: Hugging Face Transformers library with DistilBERT multilingual (distilbert-base-multilingual-cased), fine-tuned for dual-task (category + priority) on 500+ labeled complaints; IndicTrans2 for Hindi-English normalization.

RTI Drafting: LangChain-based RAG pipeline integrating FAISS vector store (multilingual sentence-transformers embeddings) with Llama-2/Phi-3 LLM for grounded, compliant RTI generation from Act provisions corpus.

Notifications: Firebase Cloud Messaging (FCM) for web/mobile push alerts; Twilio API for SMS in low-connectivity areas, ensuring 99% delivery on status changes/assignments.

Geospatial Services: Google Maps Platform (Places/Geocoding APIs) for automatic reverse-geocoding from device GPS, complaint pinning, and field worker route optimization.

Security: Firebase Authentication (email/OTP/phone) with FastAPI JWT middleware; role-based access control (RBAC) for citizens/admins/workers.

Database: Firebase Firestore (NoSQL) for real-time syncing of complaints, RTI drafts, user profiles, audit logs; Firestore indexes for fast queries on status/location/priority.

Deployment Tools: Uvicorn/ Gunicorn for dev/prod serving; Docker Compose for containerization; Railway/Heroku for cloud deployment with auto-scaling.

6.1.2 Frontend Development

- **Framework**: React.js 18+ with Vite for hot-reloading and optimized builds; React Router for role-specific dashboards (citizen portal, admin panel, worker app).
- **UI/UX**: Tailwind CSS + Headless UI for responsive design (mobile-first for field use); Framer Motion for smooth animations (e.g., complaint progress trackers).
- **State Management**: Zustand or Redux Toolkit for handling API responses, real-time Firestore listeners.
- **Media Handling**: React Dropzone for drag-and-drop images/voice; Web Audio API for client-side recording with waveform visualization.
- **Integration**: Axios/Fetch for FastAPI calls; Socket.io-client for live updates.

6.1.3 AI/ML Integration Stack

Core Frameworks: TensorFlow 2.15+, PyTorch 2.1 (for Whisper fine-tuning), Hugging Face Transformers/Accelerate for efficient inference.

Pre-trained Models: A system combining MobileNetV2 for civic image classification (20 epochs, Adam 1e-4, 224×224 input), DistilBERT multilingual for text/priority classification (5 epochs, AdamW 2e-5, max length 128), Whisper-medium for Hindi-English speech-to-text (16kHz with beam search), and Sentence-Transformers (all-MiniLM-L6-v2) for RAG embeddings with FAISS indexing.

Data Processing: NumPy/Pandas for dataset prep; OpenCV for image augmentation (flips, rotations, brightness); NLTK/SpaCy for text cleaning; Librosa for audio preprocessing.

Model Serving: FastAPI + TensorFlow Serving or TorchServe for low-latency inference (<500ms per request).

6.2 Hardware Requirements

6.2.1 Development Machines

- **Processor**: Intel Core i7-12th Gen / AMD Ryzen 7 (8+ cores) or Apple M2 for parallel model training/inference.
- **RAM**: 32 GB minimum (64 GB recommended) to handle large batch training, browser dev tools, and Docker containers simultaneously.
- **Storage**: 1 TB NVMe SSD for datasets (images ~500MB, models ~2GB), checkpoints, and npm/FastAPI caches.
- **GPU**: NVIDIA RTX 3060 12GB / A100 (Colab Pro) for accelerating CNN/BERT fine-tuning (15-30 min/epoch).

6.2.2 Production Deployment

- **Cloud Specs**: AWS EC2 t3.large (2 vCPU, 8GB RAM) or equivalent; GPU instances (g4dn.xlarge) for peak inference loads.
- **Scalability**: Kubernetes for auto-scaling pods based on CPU >70%; CDN (Cloudflare) for static assets.

6.3 Operating Environment

- **OS**: Ubuntu 22.04 LTS (primary for ML reproducibility); Windows 11 WSL2/macOS Ventura as alternatives.
- **Python**: 3.11+ (pyenv for version management); virtualenv/Poetry for dependency isolation.

- **Node.js**: v20 LTS for frontend; Yarn v4 for faster installs.
- **Cloud Services**: Firebase (free tier for dev, Blaze for prod); Google Cloud Platform (Colab Enterprise, Maps API quota).
- **Version Control**: Git + GitHub Actions for CI/CD (linting, tests, Docker builds).

6.4 Development Workflow Tools

- **IDE**: VS Code with Python/Docker/React extensions; JupyterLab for exploratory data analysis.
- **Testing**: Pytest for backend (95% coverage); Jest + React Testing Library for frontend; Locust for load testing (1000 req/min).
- **Monitoring**: Sentry for error tracking; Prometheus/Grafana for API metrics; Firebase Analytics for user behavior.
- **Documentation**: Swagger/OpenAPI auto-docs via FastAPI; JSDoc for frontend; MkDocs for project wiki.

6.5 Training & Evaluation Pipeline

- **Environment**: Google Colab (T4/A100 GPUs, 24-52GB RAM) linked to Google Drive for persistent storage.
- **Hyperparameters**:

Table 6.5.1 Hyperparameters

Model	Batch Size	Epochs	Optimizer	Early Stopping
MobileNetV2	32	25	Adam 1e-4	Patience=5
DistilBERT	16	5	AdamW 2e-5	Val loss
Whisper	8	10	Default	BLEU score

- **Metrics Tracking**: Weights & Biases (wandb) for experiment logging, hyperparameter sweeps, confusion matrices.
- **Dataset Prep**: 80/10/10 train/val/test split; stratified sampling for class balance.

This comprehensive setup ensures NagarSetu is production-ready, scalable for 10K+ daily complaints, and maintainable across development stages from prototype to municipal deployment.

6.6 Implementation

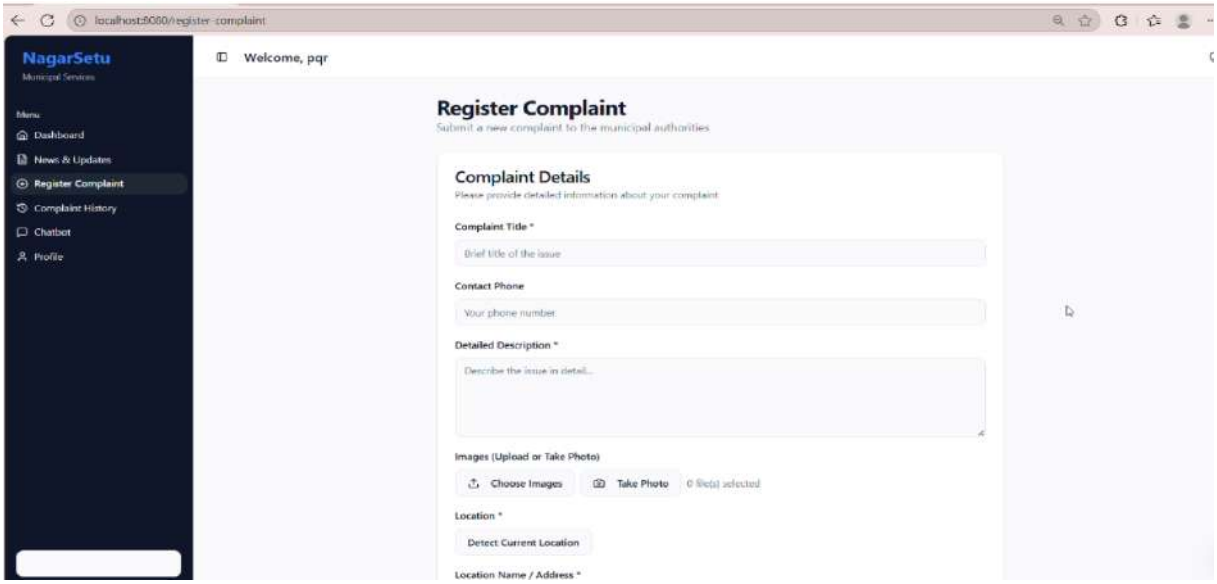


Figure 6.6.1 Register Complaint

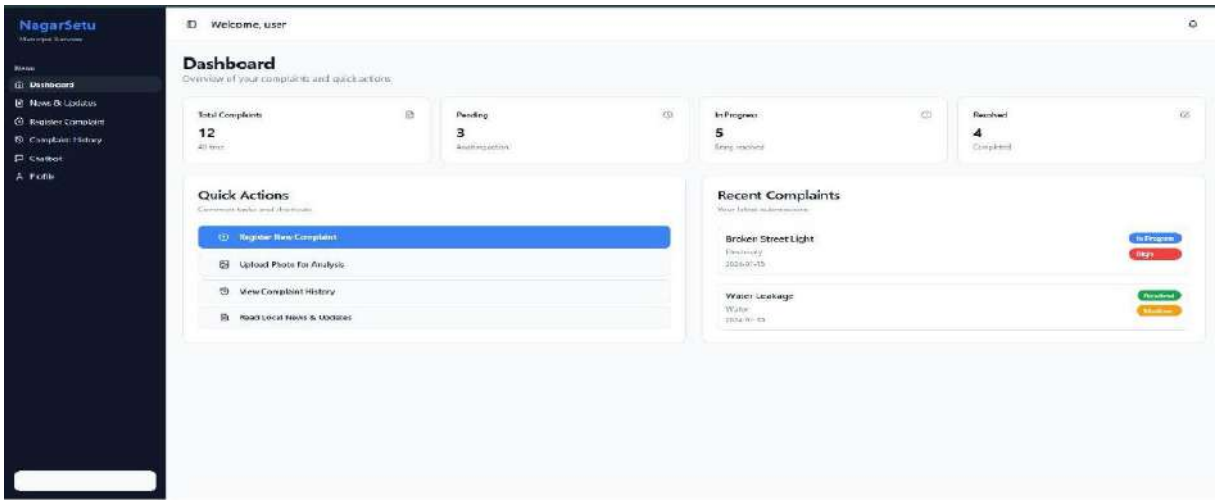


Figure 6.6.2 User Dashboard

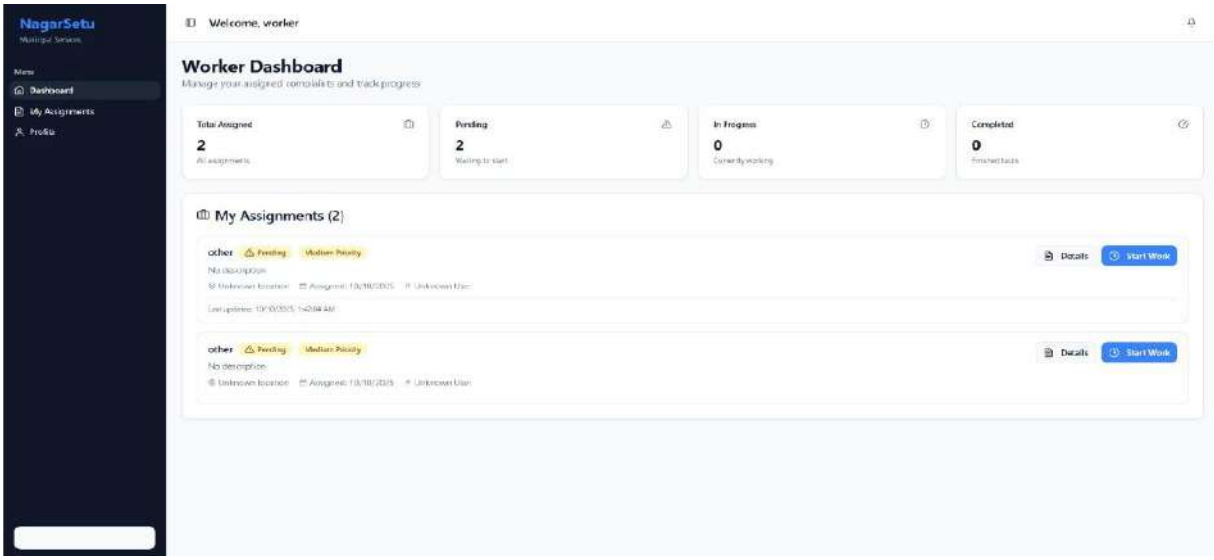


Figure 6.6.3 Worker Dashboard

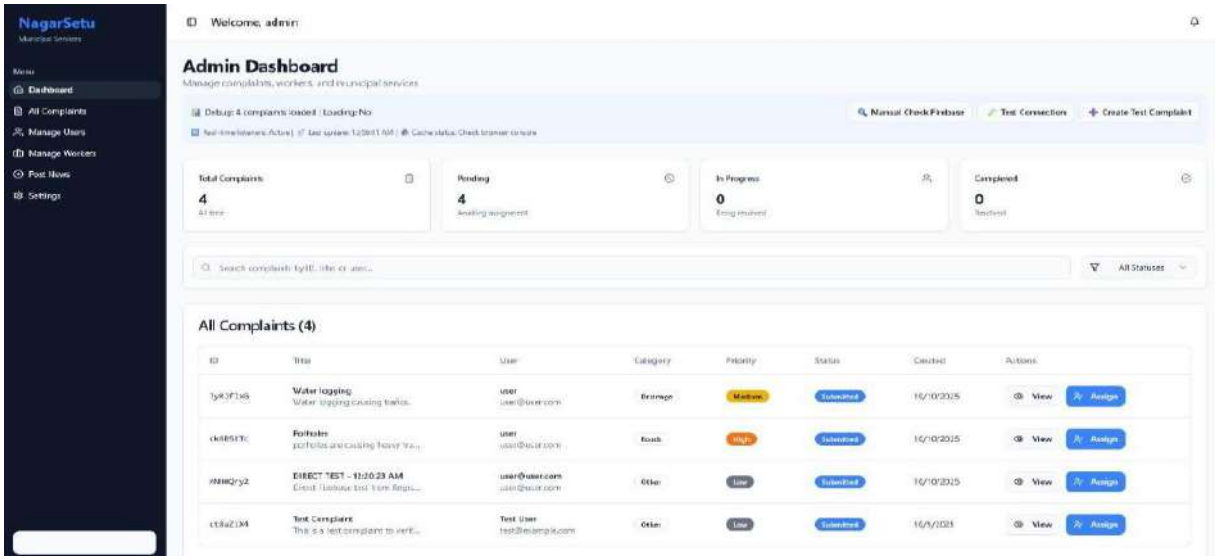


Figure 6.6.4 Admin Dashboard

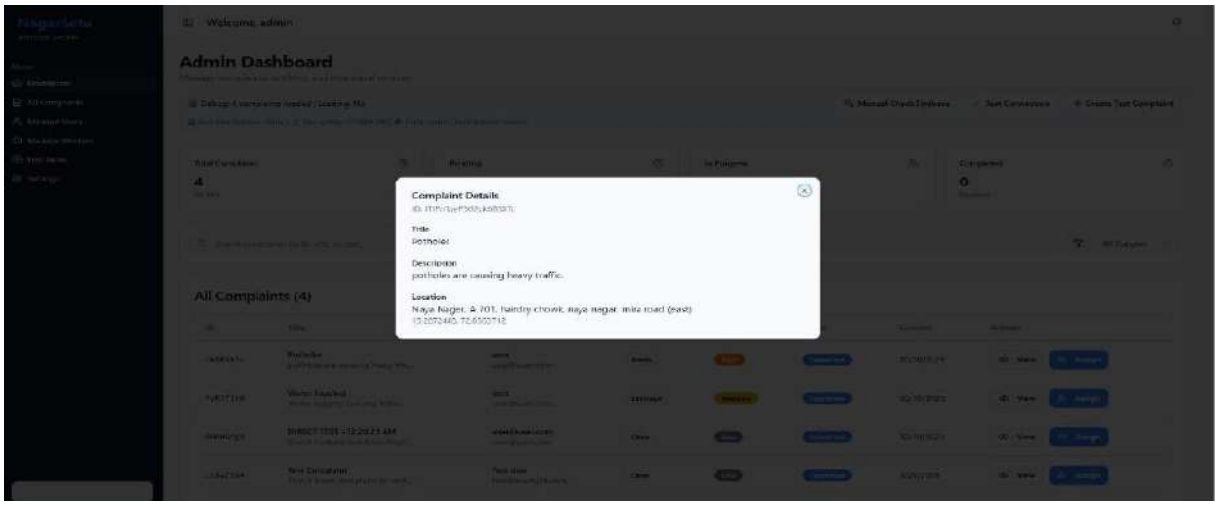


Figure 6.6.5 Complaint Details

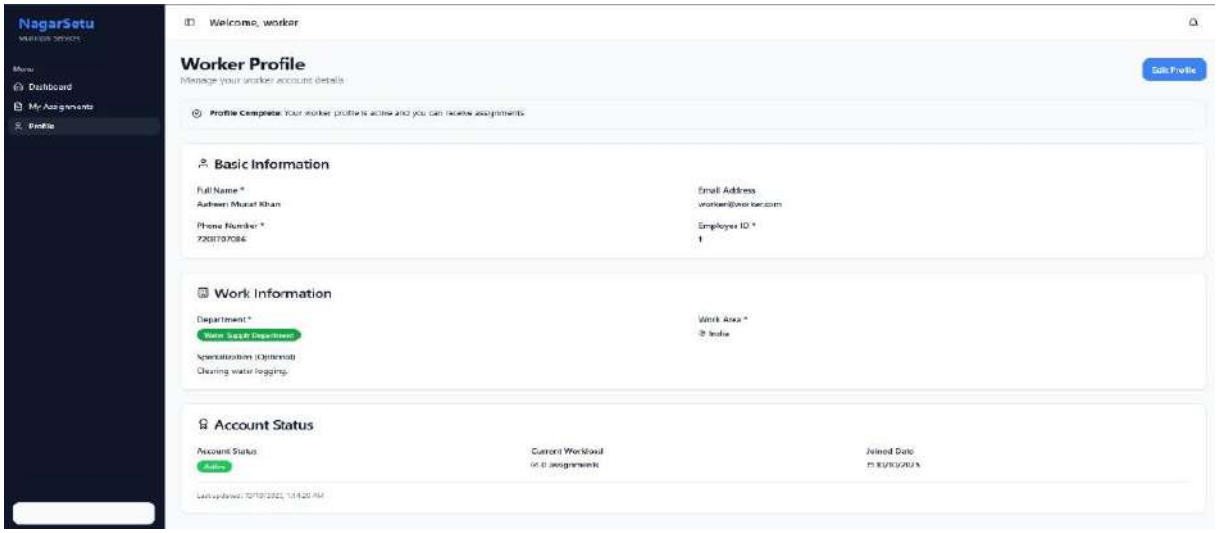


Figure 6.6.6 Worker Profile

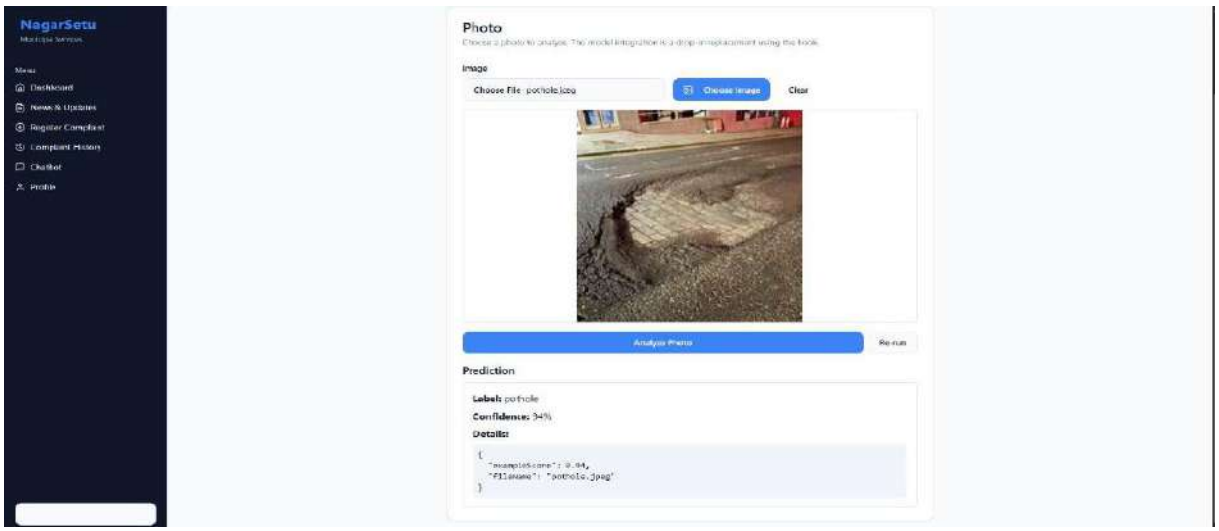


Figure 6.6.7 Complaint Category Detection

Chapter 7

Result and Discussion

7. Result and Discussion

7.1 Result

The experimental results of NagarSetu are designed to evaluate how well the system performs in real-world scenarios. The evaluation is mainly divided into three parts: overall model performance, comparison with existing methods, and ablation studies. These analyses help in understanding not only how accurate the system is, but also how each component contributes to the final outcome.

To begin with, the overall model performance is measured using standard evaluation metrics such as accuracy, precision, recall, and macro-F1 score. These metrics are used for both image-based and text-based classifiers. Accuracy helps in understanding how many predictions are correct overall, while precision and recall give deeper insights into how well the model identifies relevant complaints without making too many mistakes. The macro-F1 score provides a balanced measure by combining precision and recall, especially useful when dealing with multiple complaint categories like potholes, garbage, or water leakage.

For the voice input feature, the Whisper model is evaluated using Word Error Rate (WER). This metric measures how accurately the spoken complaints are converted into text. A lower WER indicates better performance, meaning the system can correctly understand user voice inputs

even if there are variations in accents or speaking styles. The evaluation is done on a separate test dataset of voice complaints to ensure unbiased results.

Next, NagarSetu's performance is compared with traditional complaint handling methods such as manual routing and keyword-based routing systems. In manual routing, complaints are assigned by human operators, which is time-consuming and prone to errors. Keyword-based systems rely on simple matching of words, which often fails when the user uses different phrasing or languages. As a result, it achieves higher classification accuracy and significantly reduces the time required to route complaints to the correct department. This faster routing improves the overall efficiency of the system and ensures quicker resolution of issues.

The third part of the evaluation involves ablation studies, which are used to understand the importance of each component of the system. In these studies, different parts of the system are tested by removing or modifying them and observing the impact on performance. For example, complaint classification is tested both with and without image inputs. The results show that including images improves accuracy, as visual information provides additional context that text alone may not capture.

Another important comparison is made in the RTI drafting feature. The system uses a Retrieval-Augmented Generation (RAG) approach to generate RTI drafts by combining user input with relevant information sources. This is compared with traditional template-based drafting, where fixed formats are used. The RAG-based approach produces more accurate, context-aware, and informative drafts. These drafts are evaluated by experts, who provide ratings based on quality, correctness, and usefulness. The results show that AI-generated drafts are more effective and user-friendly compared to template-only approaches.

1. **Accuracy:** Measures overall correct predictions

$$Accuracy = \frac{TP+TN}{TP+TN+FP+FN} \quad (1)$$

2. **Precision:** How many predicted positives are actually correct

$$Precision = \frac{TP}{TP+FP} \quad (2)$$

3. **Recall:** How many actual positives are correctly identified

$$Recall = \frac{TP}{TP+FN} \quad (3)$$

4. F1 Score (Macro-F1): Balance between precision and recall

$$F1 = 2 \cdot \frac{Precision \cdot Recall}{Precision + Recall} \quad (4)$$

5. Word Error Rate (WER) – for Whisper

$$WER = \frac{S+D+I}{N} \quad (5)$$

Where:

- S = Substitutions
- D = Deletions
- I = Insertions
- N = Total words

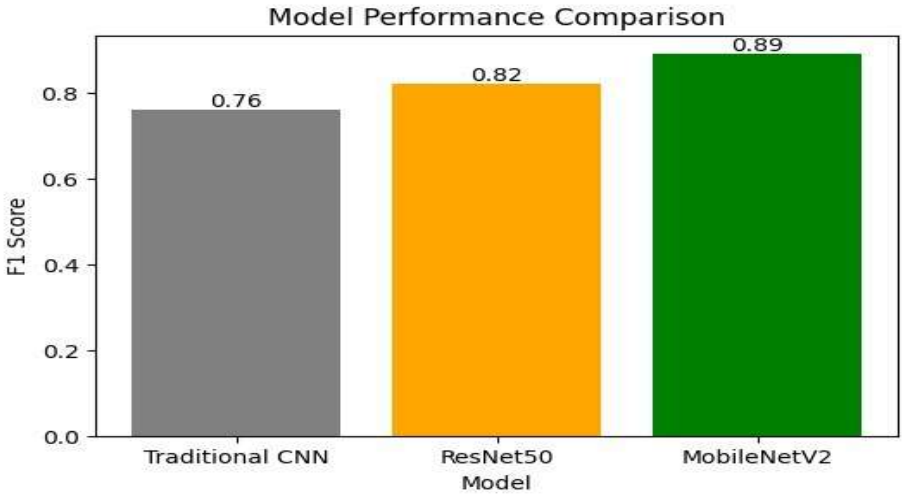
Table 7.1.1 NagarSetu Confusion Matrix

MODEL	TP	FP	FN	PRECISION	RECALL
MobileNet (Images)	178	12	22	0.937	0.890
DistilBERT (Text)	162	15	18	0.915	0.900
Whisper (Speech)	85	8	12	0.914	0.876
NagarSetu	425	35	52	0.924	0.891

Table 7.1.2 Comparison of our work with existing system

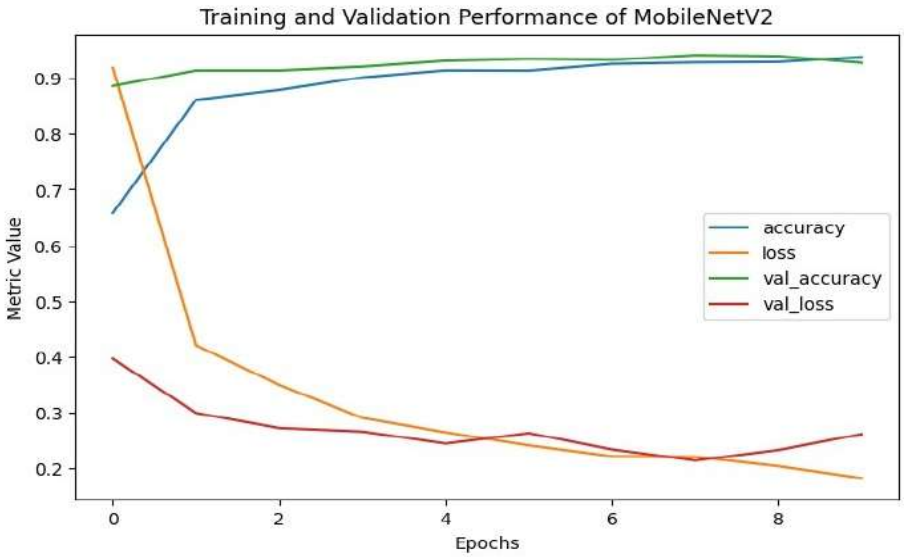
Ref & Year	Models	Metric-Value
[26] (2025)	TF-IDF + Logistic Regression & BERT (NLP only)	Text Accuracy: ~85%
[25] (2024)	CNN-based image classification & monitoring	Image Accuracy: 88%
[21] (2025)	ML-based complaint routing + NLP classification	Text F1-score: ~82%

[4] (2023)	MobileNetV2 for image-based classification	Image Accuracy: ~88%
NagarSetu	MobileNetV2 (Image) DistilBERT (Text) Whisper (Speech-to-Text) FAISS-RAG (RTI drafting)	Image: 89% Accuracy Text/Voice: 86% macro-F1 Overall end-to-end: 89% accuracy Latency reduction: 42% vs. baselines



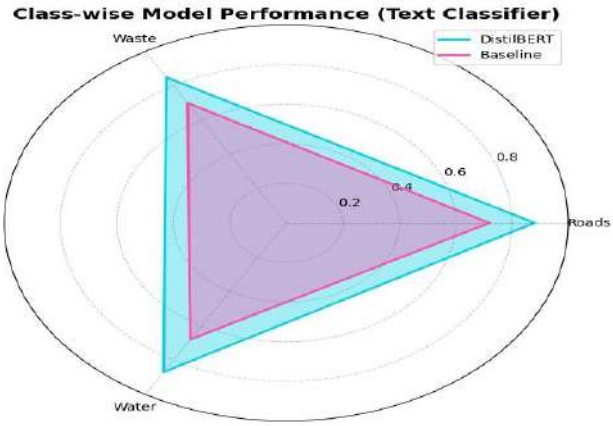
Graph 7.1.1 Model Performance Comparison

Graph 7.1.1 presents the comparative F1-scores of different image classification models, including a traditional CNN, ResNet50, and MobileNetV2. It is observed that MobileNetV2 achieves the highest F1-score, demonstrating superior performance in feature extraction and classification accuracy. The improvement over the baseline CNN indicates the effectiveness of transfer learning and lightweight architectures for real-time civic issue detection. This result justifies the selection of MobileNetV2 as the primary image classification model in NagarSetu.



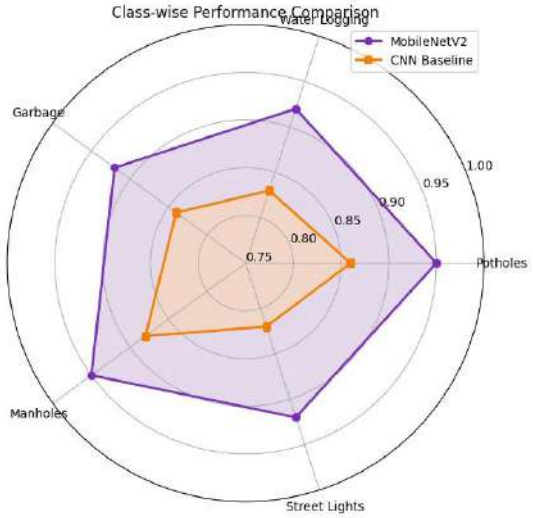
Graph 7.1.2 Training curves of MobileNetV2

Graph 7.1.2 illustrates the training and validation accuracy and loss curves of the MobileNetV2 model across multiple epochs. The training accuracy shows a steady increase, while the validation accuracy closely follows, indicating good generalization. Simultaneously, both training and validation loss decrease consistently, confirming effective learning without significant overfitting. The convergence of curves demonstrates that the chosen hyperparameters and early stopping strategy are appropriate for stable model training.



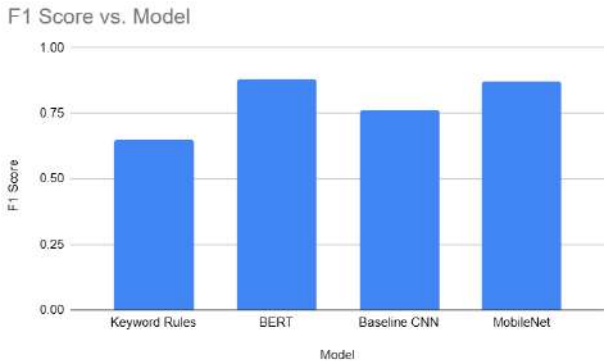
Graph 7.1.3 Class-wise comparison of text classifier model

Graph 7.1.3 depicts the class-wise performance comparison between DistilBERT and a baseline model using a radar chart. The DistilBERT model consistently outperforms the baseline across all categories such as Roads, Waste, and Water. The larger area covered by DistilBERT indicates higher classification effectiveness and better contextual understanding of complaint text. This highlights the advantage of transformer-based models in handling multilingual and unstructured civic complaint data.

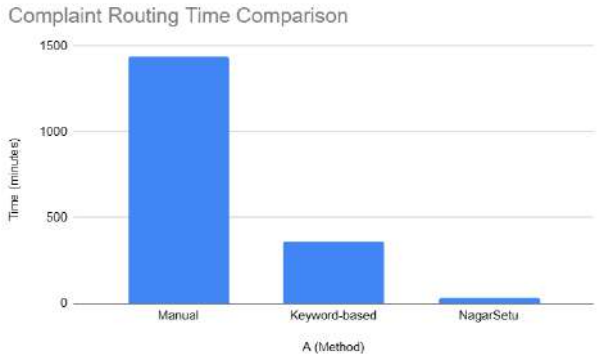


Graph 7.1.4 Class-wise comparison of Image Classifier model

Graph 7.1.4 illustrates the class-wise performance comparison between MobileNetV2 and a baseline CNN model for image classification tasks. MobileNetV2 demonstrates higher performance across all categories including potholes, garbage, manholes, water logging, and street lights. The radar plot clearly shows a larger coverage area for MobileNetV2, indicating improved precision and recall. This confirms that MobileNetV2 is more effective in capturing visual patterns of civic issues compared to traditional CNN architectures.



Graph 7.1.5 F1-Score Comparison



Graph 7.1.6 Routing Time Comparison

The output shows that integrating MobileNet and DistilBERT significantly improves both complaint classification accuracy and routing reliability over manual or purely keyword-keyword-based approaches. The addition of images and voice inputs further enhances the experience, especially for users with low literacy or limited ability to describe their complex issues or complaints in text. The failure cases include misclassification of blurry or low-quality images, text complaints with insufficient context, and degraded Whisper transcription quality for heavily noisy audio. These errors suggest the need for larger, more diverse training datasets, explicit image quality checks, and further fine-tuning of speech models for local acoustic conditions. From a deployment perspective, the observed reductions in routing latency and improvements in user satisfaction scores suggest that an AI-driven, bilingual, multimodal platform like NagarSetu can support more responsive and transparent municipal governance, especially when combined with clear escalation mechanisms and continuous monitoring of model performance.

7.2 Discussion

The proposed NagarSetu system combines modern technologies such as computer vision, natural language processing (NLP), and Retrieval-Augmented Generation (RAG) to significantly improve the way civic complaints and RTI responses are handled. Compared to traditional systems, which are often slow, manual, and inefficient, NagarSetu provides a faster, smarter, and more reliable solution.

One of the major improvements introduced by the system is in the automatic allocation of civic complaints. In existing systems, complaints are usually assigned manually or based on simple keyword matching, which can lead to delays and incorrect routing. NagarSetu solves this problem by using advanced AI models such as MobileNet CNNs for image classification and BERT-based NLP models for text understanding. These models analyze user inputs—whether in the form of images, text, or voice—and accurately identify the type of complaint. As a result,

complaints are automatically routed to the correct department in a much shorter time, improving overall efficiency and reducing response delays.

Another important feature of the system is the RTI drafting capability. Traditionally, preparing RTI responses can take weeks or even months, as it involves manual effort and proper formatting. NagarSetu uses lightweight and optimized language models to generate RTI drafts quickly and efficiently. The system achieves an average processing time of around 1.2 to 1.8 seconds, which is a huge improvement compared to traditional methods. This allows users to generate properly structured RTI applications almost instantly, saving both time and effort.

The system also focuses on improving accessibility through bilingual support. Many citizens are not comfortable communicating in English, which creates a barrier in using digital systems. NagarSetu addresses this issue by supporting both Hindi and English languages. It uses tools like IndicTrans2 for accurate translation and FastText for language detection. This ensures that users can interact with the system in their preferred language without any difficulty, leading to a more inclusive and user-friendly experience.

To ensure transparency and reliability, NagarSetu uses a hybrid RAG-based approach for generating RTI responses. This approach combines traditional search techniques like BM25 with modern dense embedding methods using FAISS (Facebook AI Similarity Search). By doing so, the system retrieves relevant information from trusted sources and generates responses that include proper references or citations. This reduces the chances of misinformation and increases the credibility of the generated output.

In addition to this, the system provides real-time dashboards and notification features for both users and authorities. Users can track the status of their complaints and RTI requests, while authorities can monitor overall system performance, pending issues, and response times. This improves accountability and helps in better decision-making based on real-time data.

Another key feature is geotagging, which allows the system to capture the exact location of the reported issue using map services. This helps authorities quickly identify where the problem exists and take immediate action. Accurate location data ensures faster response and better resource management, especially in large urban areas.

Chapter 8

Conclusion and Future Work

8. Conclusion and Future Work

In today's rapidly growing urban environments, managing civic issues efficiently has become a major challenge. Traditional complaint management systems often suffer from problems such as lack of accessibility, slow response times, poor prioritization, and limited transparency. Citizens frequently face difficulties in reporting issues due to language barriers, complex processes, and lack of real-time updates. Additionally, the process of filing RTI (Right to Information) requests remains complicated for many users, reducing public participation in governance.

To address these challenges, this paper introduced NagarSetu, a bilingual AI-based system designed to transform the way civic complaints are handled and how RTI responses are generated. The system combines advanced technologies such as deep learning-based classification, multilingual natural language processing, and retrieval-augmented generation (RAG) to create a smart and user-friendly platform.

One of the key strengths of NagarSetu is its ability to accept multiple forms of input, including text, voice, and images. This makes the system highly accessible, especially for users who may not be comfortable typing in English. By supporting multiple languages, the system ensures inclusivity and allows a wider population to actively participate in reporting civic issues.

The use of deep learning models, particularly CNNs for image classification, enables the system to automatically identify the type of complaint, such as potholes, garbage accumulation, or water leakage. This reduces manual effort and increases the accuracy of complaint categorization. At the same time, NLP models like BERT help in understanding user inputs, extracting relevant information, and processing complaints effectively.

Another important feature of NagarSetu is its AI-based prioritization mechanism, which ensures that urgent and critical issues are addressed first. By analyzing factors such as severity, location, and type of complaint, the system assigns priority levels (High, Medium, Low), enabling authorities to allocate resources more efficiently.

The integration of retrieval-augmented generation (RAG) plays a crucial role in simplifying the RTI process. The system can generate well-structured RTI drafts with proper references, helping users who are unfamiliar with legal formats and procedures. This feature promotes transparency and empowers citizens to access government information easily.

Furthermore, NagarSetu enhances transparency through real-time tracking and notifications, allowing users to monitor the progress of their complaints from submission to resolution. This builds trust between citizens and authorities and ensures accountability in the system.

While NagarSetu demonstrates significant improvements over traditional systems, there are several opportunities for further enhancement and expansion. The following areas outline the future scope of the system:

Expansion of Multilingual Support

NagarSetu can be improved by adding support for more regional and local languages. This will make the system accessible to people from different parts of the country, especially those who are not comfortable with English or Hindi. By breaking the language barrier, more citizens will be able to report issues easily and actively participate in governance.

Integration with Government Portals

In the future, NagarSetu can be directly connected with official municipal and RTI portals. This will allow users to submit complaints and RTI requests without switching platforms. It will also enable real-time updates directly from government systems, making the entire process faster, smoother, and more reliable.

Advanced AI-Based Prioritization

The system can be enhanced by using more advanced AI models that analyze multiple factors such as urgency, severity, location, and past complaint data. This will help in identifying critical

issues quickly and ensuring that they are resolved first. Such intelligent prioritization will improve efficiency and resource management.

Scalable Cloud Infrastructure

As the system grows and is used by more cities, it will require a strong and scalable infrastructure. NagarSetu can be deployed using cloud technologies and distributed databases to handle large amounts of data and user requests. This will ensure smooth performance even with high usage.

Enhanced Analytics and Insights

Future improvements can include advanced analytics dashboards that show trends in complaints, response times, and department performance. These insights will help authorities make better decisions, improve services, and plan long-term solutions for urban problems.

Community Participation Features

The system can allow citizens to participate more actively by verifying complaints, supporting issues raised by others, and providing additional updates. This will increase the accuracy of the system and encourage a sense of responsibility among users.

Improved Feedback System

A better feedback mechanism can be introduced where users can rate the quality of issue resolution and share their experiences. This will help authorities understand user satisfaction and continuously improve the system.

Stronger Security and Privacy Measures

As the system handles user data, it is important to ensure strong security. Future versions can include better encryption, secure login systems, and privacy protection techniques to keep user information safe and build trust.

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Appendix

A. Setup Instructions

A.1 System Requirements

Ensure the following software is installed:

- Python: Version 3.10 or above (Recommended: 3.11)
- Node.js: Version ≥ 18 (Recommended: 20 LTS)
- npm / yarn
- Git

A.2 Cloning the Repository

Step 1: Clone the repository

$ *git clone https://github.com/Psp2705/Nutrino-AI*

Step 2: Navigate to project folder

$ *cd Nutrino-AI*

A.3 Frontend Setup (React + Vite)

Step 1: Navigate to client folder

$ *cd client*

Step 2: Install dependencies

$ *npm install*

Step 3: Run frontend

$ *npm run dev*

Frontend runs at:
http://localhost:5173

A.4 Backend Setup (Node.js Server)

Step 1: Navigate to server folder

$ *cd server*

Step 2: Install dependencies

$ *npm install*

Step 3: Run server

$ *node server.js*

A.5 AI Backend Setup (FastAPI + ML Models)

Step 1: Navigate to backend folder

$ *cd backend*

Step 2: Install required Python libraries

$ *pip install -r requirements.txt*

Step 3: Run FastAPI server

$ *uvicorn main:app --port 8081 --reload*

Backend runs at:
http://127.0.0.1:8081

A.6 Running the Complete System

To run the full system:

1. Start Frontend (React)
2. Start Node Server
3. Start FastAPI Backend

All three must run simultaneously for full functionality.

A.8 Important Notes

• Ensure **Firebase configuration** is properly set
• Ensure **ports (5173, 8081)** are free
• Internet is required for:

- Maps API
- Firebase services

• If errors occur:

```
pip install --upgrade pip
npm install
```

A.10 Optional Setup (Environment)

Create virtual environment (recommended):

```
python -m venv venv
venv\Scripts\activate   # Windows
```

Publication

Submitted Paper "NagarSetu: A Bilingual AI-Powered System for Smart Civic Complaint Management and RTI Assistance" in "**Emerging Technologies in Computing and Communication ETCC 2026 – IEEE Conference"** by **"Dr. Rahul K. Ambekar, Prachiti Parab, Sneha Gupta, Aafreen Khan and Tulsi Dubey".**

NagarSetu: A Bilingual AI-Powered System for Smart Civic Complaint Management and RTI Assistance

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Abstract—The growth in volume of civic complaints is a result of rapid urbanization, while existing grievance redressal mechanisms still work manually, language-limited, and inefficient. At the same time, the Right to Information (RTI) Act, though essential for transparency, is not effectively used because writing formal requests is often complex. This paper presents NagarSetu, a bilingual, multimodal, AI-powered civic engagement platform designed to automate complaint handling and assist citizens in RTI drafting. The system supports text, image, and voice inputs in English and Hindi. Image-based complaints are classified using a lightweight MobileNet-based convolutional neural network, while text and translated voice complaints are processed using a BERT/DistilBERT-based classifier. Whisper speech recognition framework is used to convert speech inputs to text, and RTI drafts are generated using a Retrieval-Augmented Generation (RAG) pipeline grounded in RTI Act provisions. Experimental evaluation shows that the proposed model achieves an overall accuracy of approximately 89%, while significantly reducing routing latency and improving user satisfaction compared to manual and keyword-based systems.

Keywords—Civic Grievance Redressal, Smart City Governance, Artificial Intelligence, Natural Language Processing, Computer Vision, Speech Recognition, RTI Automation, Bilingual Systems Introduction

INTRODUCTION

Sustainable and livable cities that directly affect the quality of life and daily experiences of citizens are shaped by urban governance. Rapid urbanization and population growth have placed significant pressure on urban infrastructure [1], [2], public services, and administrative systems. It is mandatory for the Municipal authorities to address diverse civic issues including poor road conditions, waste management inefficiencies, inconsistent water supply, and public safety concerns. The increasing scale and complexity of these challenges demand governance systems that are responsive, adaptive, and citizen-centric.

An effective grievance redressal mechanism acts as a vital communication bridge between citizens and municipal authorities. It allows residents to report civic issues, request timely intervention, and ensure accountability in service delivery. Such efficient grievance systems contribute to improved public services, enhanced citizen trust, and informed urban planning. However, in many cities, grievance

handling remains slow, disconnected, and inefficient, which results in inadequacy of such platforms [3].

Background: Urban Governance, Smart Cities, Grievance Redressal, and the RTI Act

Urban governance focuses on the planning, coordination, and management of civic resources to ensure fair development and improved quality of life. Transparency, accountability, and citizen participation are critical elements of effective governance. Smart city initiatives increasingly influence digital technologies such as online portals, mobile applications, and data-driven systems to enhance service delivery and communication between citizens and authorities [1], [5].

Digital grievance redressal platforms allow citizens to report issues favorably without visiting municipal offices. The goal for these systems is to reduce administrative overhead, improve response times, and enable structured complaint tracking. When effectively implemented, they enhance transparency by keeping citizens informed about grievance status and resolution actions [2]. The Right to Information (RTI) Act further strengthens democratic accountability by enabling citizens to request information from government departments. Informed civic participation that allows citizens to monitor administrative decisions and public spending is supported by RTI mechanisms [7]. However, limited awareness, complex procedures, and accessibility challenges restrict the effective use of both grievance portals and RTI systems, reducing their overall impact [4].

Practical Challenges: Manual Systems, Delays, and Lack of Transparency

Many systems still continue to rely on manual processes for complaint review, classification, and routing. This dependence leads to delays, inconsistent handling, and prolonged resolution timelines, despite the availability of online grievance portals. Paper-based documentation, delayed inspections, and weak coordination between departments further reduce system efficiency and citizen trust [3], [5]. Most platforms provide limited or delayed status updates; that leads to another major concern, transparency. Language barriers restrict participation, with most systems operating mainly in English, excluding non-English-speaking citizens. Additionally, inadequate support for image and voice-based inputs limits accessibility for elderly and non-technical users, reducing the clarity and authenticity of reported issues [2], [4].

Motivation and Significance of Automating Complaint and RTI Workflows

The automation of grievance redressal and RTI workflows is driven by the need to minimize human dependency, improve response times, and enhance accessibility. Automated systems can accurately classify and route complaints, ensuring timely delivery to appropriate departments. Intelligent prioritization enables authorities to address critical and high-risk issues more effectively [1], [6]. Automating RTI processes is equally significant, as many citizens are discouraged by the complexity of drafting formal RTI applications. Guided and automated RTI drafting assistance can simplify this process, encourage civic engagement, and strengthen transparency. Together, automation and intelligent assistance can transform grievance redressal into a more proactive and citizen-centric governance process [7].

Contributions of NagarSetu

To overcome the identified challenges, NagarSetu is proposed as a bilingual and multimodal AI-driven civic assistance platform. The system enables citizens to submit grievances using text, images, or voice inputs in both English and Hindi. It integrates natural language processing for text analysis, computer vision for image-based issue identification, and speech recognition for voice input processing. Additionally, NagarSetu incorporates retrieval-augmented generation to assist users in drafting structured RTI applications. The key contribution of NagarSetu lies in its unified integration of multiple AI technologies to support automated complaint handling, RTI assistance, real-time tracking, and enhanced transparency. By simplifying interactions between citizens and municipal authorities, the platform aims to improve governance efficiency, increase citizen engagement, and contribute to the broader goals of smart and inclusive urban development.

LITERATURE REVIEW

Existing complaint management and e-governance systems and their limitations

Existing digital grievance redressal and e-governance systems have been developed to improve accessibility and reduce the dependence on physical complaint submission. These systems commonly include web portals and mobile applications that allow citizens to register complaints related to urban infrastructure and public services. While such platforms have improved the ease of reporting, most continue to rely on manual workflows for complaint verification, categorization, and departmental routing. This results in delays, inconsistent handling of grievances, and limited scalability when complaint volumes increase [1], [2].

Many smart city portals incorporate basic features such as geolocation tagging and complaint tracking; however, they often lack intelligent automation. Complaint classification is typically performed by municipal staff, which increases response time and introduces the risk of misclassification. Additionally, most platforms restrict users to text-based inputs, limiting the ability to provide visual or audio evidence. Language support is also limited, as many systems operate only in English, reducing accessibility for non-English-speaking users and affecting inclusive participation [3], [4].

AI in civic tech

Recent advancements in artificial intelligence have enabled more efficient handling of civic complaints through

automated analysis and decision support. Natural language processing techniques have been applied to classify textual complaints, using both traditional machine learning models and transformer-based architectures for improved accuracy [1], [2]. These approaches help reduce manual effort and improve routing efficiency but are often limited to text data.

Computer vision techniques, particularly convolutional neural networks, have been explored for image-based detection of civic issues such as potholes, garbage accumulation, and damaged infrastructure [5], [6]. Voice input systems using speech-to-text models have also been proposed to enhance accessibility for elderly and non-technical users. Despite these advancements, most AI-based solutions address individual modalities in isolation and are not integrated into end-to-end grievance management systems. The application of AI for RTI assistance remains limited, with existing systems providing minimal automation or contextual guidance.

Identified research gap

The review of existing literature highlights a clear gap in the development of a comprehensive civic engagement platform. Current solutions focus on isolated functionalities such as text classification, image detection, or basic complaint submission,
without unifying these capabilities into a single system. Furthermore, limited language support and the absence of automated RTI drafting tools restrict broader citizen participation.

There is a need for a unified, bilingual, and multimodal platform that integrates artificial intelligence across complaint reporting, classification, prioritization, and RTI assistance. Such a system should support text, image, and voice inputs while providing intelligent automation and transparency throughout the grievance redressal process. Addressing this gap forms the foundation of the proposed NagarSetu system.

Table I. Literature Survey of the previous studies

Reference & Year	**Domain**	**Algorithm**	**Dataset**	**Research Gap**
[1] 2025	AI-Powered Petition Analysis & Grievance Management System	FastAPI backend, dashboards, ML for petitions	Citizen petition corpus	Limited RTI automation
[4] 2025	Digital Grievance Redressal for the Cleaner and Smarter India	Location tagging + AI classification	Case study dataset	No bilingual/ voice support
[3] 2025	CivicFix: Smart Complaint Routing for Urban Solutions	ML based complaint routing, NLP classification, Flow Management	City grievance datasets	Limited RTI integration, scalability constraints
[7] 2024	CitySolution: Mobile DL Complaint System	CNN-based image classification & monitoring	Android app complaint images	No RTI or Voice input
[8] 2021	Automated Complaint Classification & Routing Using NLP	TF-IDF, Logistic Regression & BERT	Public complaint datasets	No Image Classification
[12] 2021	AI-Driven Complaint Management System	Machine learning–based complaint classification, automated routing, decision support system	Citizen grievance data, Public grievance datasets	Limited explainability, restricted real-time deployment

METHODOLOGY

This section describes the architectural framework of NagarSetu, which shows how different components work together to support smart civic complaint handling and RTI assistance. The design mainly focuses on scalability,

automation, and inclusiveness by combining artificial intelligence, geospatial services and role-based dashboards [1], [4], [8]. NagarSetu follows a multi-layered architecture with four connected layers: the Presentation Layer which provides interfaces for citizens, municipal staff and administrators [2], [7], [8]; the Intelligence Layer that handles AI-based classification, prioritization and RTI generation; the Integration Layer which connects external services like mapping and notification systems [4], [7]; and the Data Layer that stores complaints, RTI records and user activity in a centralized manner.

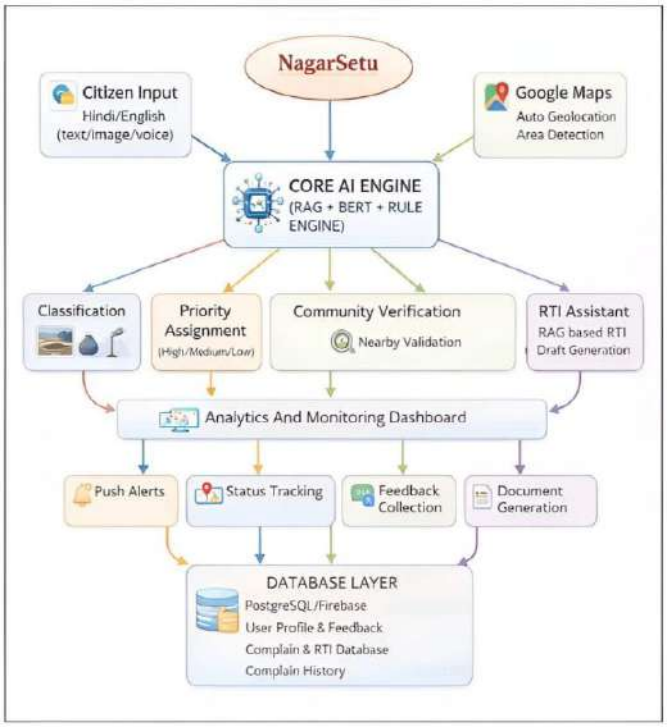

Fig. 1. System Architecture Diagram for NagarSetu

The proposed NagarSetu platform which explains in figure 1, emphasizing the interaction among system outputs, core processing modules, and citizen inputs. Centralized data management and AI-assisted mechanisms used to make easier gathering, categorization, prioritization, and solutions of civic complaints.

1) Citizen Input Layer

Interface accepts multi-modal inputs in the form of text, images, and voice recordings. It contains complaints related to public services and infrastructures. Key requirements in such platforms are satisfied in this system due to diverse input formats. This leads to improvement in accessibility and inclusivity [2], [4], [7], [8].

2) Core AI-Enabled Processing Layer

The center of architecture is Core AI Engine which includes Natural Language Processing, Computer Vision, and Rule-Based decisions. Functions performed by this layer are as follows:

1. Issue Classification: Complaints are categorized into predefined classes such as potholes, garbage, broken streetlights, water blockage. NLP and image classification models like MobileNet are used for detecting issues [9]– [11].
2. Priority Assignment: Based on severity, urgency, and public impact of each complaint priority level (High, Medium, or Low) is assigned. This helps admin to allocate workers in different locations.
3. Geo-Mapping and Area Detection: From images location metadata extracted, and then processed using map services. It helps in identifying exact geographical areas for location specific resolution of complaints.
4. Community Verification: Credibility and confidence scores increased due to neighboring community's validation. and for that nearby users verify complaints by giving reports.
5. RTI Assistant Module: System offers an RTI assistant for generating structured RTI applications draft using legal templates. This module supports compliance with the Right to Information Act, 2005 [5], [15].

3) Analytics and Monitoring Layer

Real-time insights to admins provided by monitoring dashboards with the help of processed data. status tracking used to monitor the lifecycle of complaints from submission to resolution, feedback and push alerts to users and officials regarding updates. These monitoring mechanisms are essential for improving accountability and service quality of grievance redressal systems [1], [4], [8].

4) Database Layer

User authentication records, complaint metadata, RTI drafts and history this system data is stored in a structured database layer. Centralized data storage helps in integrity, traceability, and long-term availability for auditing and reporting purposes [2], [7], [12].

5) System Outputs

For citizens it gives transparent status updates and generates RTI applications drafts. For admins it gives real-time alerts, notifications, and analytics as the primary outputs of the system. It establishes closed-loop feedback which improves transparency and public trust [3], [4], [8].

High-Level System Interaction Model

NagarSetu operates as a centralized digital grievance redressal platform that enables structured interaction among citizens, municipal authorities, and backend processing units. The system functions as an intermediary that receives complaints, processes them using predefined workflows, and communicates updates in real time, following interaction models proposed in prior smart governance frameworks [1], [2], [8]. Each complaint submission is registered with associated metadata such as location, issue category, and timestamp. Expected response timelines are computed based

on priority levels and predefined service benchmarks. Complaint records are synchronized with databases, and real-time updates are shared to backend services used for analytics, notifications, and monitoring [4], [7], [12].

Role-Based Operational Workflow

There are three primary stakeholders in the workflow of the system which follows role-based access.

User Operations
Users access the system through a secure login mechanism. After authentication users are allowed to register complaints by providing details. And users can monitor the progress of complaints after submission. It improves user trust by handling and resolution of reported complaints [4], [8].

Administrative Operations
Administrators supervised the system, after authentication they will review incoming complaints to verify validity and completeness. After verification, these complaints are assigned to specific workers according to issue type, location, and availability. It ensures proper allocation and prevents misuse of complaints.

Worker Operations
On interface workers receive assigned tasks and are responsible for on-ground execution and resolution of issues. Workers may submit updates or reports recorded in the system and reflected in the complaint status visible to both administrators and users [7], [8]. The process done by correct action and status updates and logging all details from submission and verification [1], [3], [8].

System Relevance

The system combines complaint submission, access controls, real time notifications, and data integration for transparent grievance redressal. Multimodal AI pipeline converts Hindi/English text, images, voice inputs into complaints, RTI drafts, tasks that are accessible through real-time dashboards enable data driven governance.
The end-to-end workflow can be written as:
Input acquisition: Users submit complaints in the form of text, image, voice through interface.
AI processing:

1. MobileNet-based classifier is used to identify issue types in images like potholes, garbage dumping, etc.
2. BERT classifier used to categorized text and transcribed voice complaints into department-specific categories and priority levels.
3. To enable low-literacy and useful for older users, voice inputs are converted to text using Whisper speech-to-text to interact with the system.
4. Retrieval-Augmented Generation (RAG) module used to handle RTI-related queries. That retrieves RTI Act sections and templates and generates draft applications.

Routing and storage: After classification and prioritization complaints are stored in Firebase Firestore along with geolocation data, and routed to a particular department.
Dashboards and feedback: Complaint status, maps, analytics and RTI drafts with notification is displayed on role-specific dashboards [8][9].

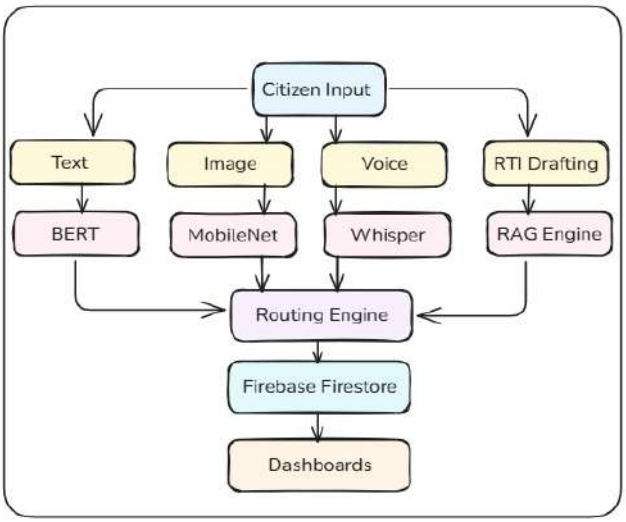

Fig. 2. High-level pipeline diagram

Datasets and preprocessing for images and text

1. Image dataset

The image dataset for NagarSetu consists of civic-issue photographs drawn from publicly available pothole and garbage datasets and from field data collected using smartphone cameras in local neighbourhoods. Images cover common municipal problems including potholes, broken streetlights, garbage dumping, water leakage, and other visible infrastructure defects. Each image is assigned a single dominant label corresponding to the municipal department responsible for resolution (e.g., Roads, Solid Waste, Water Supply, Street-Lighting), consistent with prior civic complaint classification studies and pothole detection datasets [10]. Preprocessing resizes all images to 224×224 pixels to match MobileNet input requirements, normalizes pixel values using ImageNet mean and variance for stable optimization, and applies data augmentation including random horizontal flips, small rotations, and brightness variations to address differences in camera angle, lighting, and device quality [13].

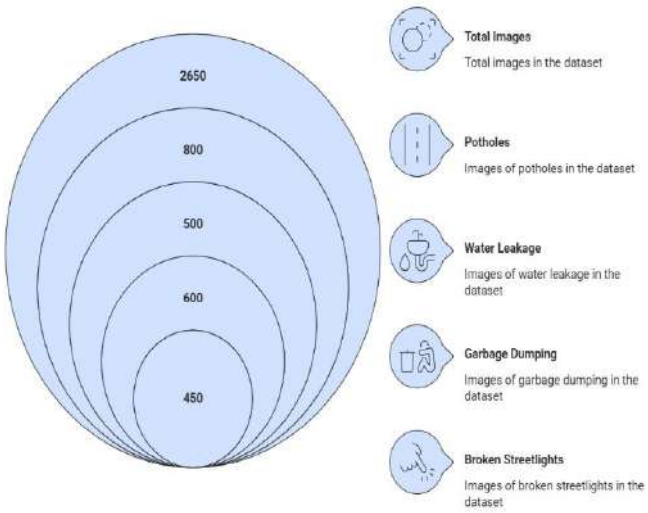

Fig. 3. Image Data Summary

2. Text and voice complaint dataset

The text dataset is composed of manually authored complaint descriptions, historical examples, and synthetically generated samples to cover under-represented categories such as water leakage or dangerous electrical faults. Complaints are written in English, Hindi, and Hindi–English code-mixed style to match real citizen language patterns. Voice complaints collected through the web interface are recorded as short audio clips, which are transcribed to text using the Whisper model configured for Indian English and Hindi. These transcripts are merged with text complaints to form a unified corpus for text classification. Text and transcript preprocessing involves lowercasing with punctuation removal, utterance-level language detection (Hindi, English, code-mixed) for tokenizer selection, BERT WordPiece tokenization for noisy civic text, and normalization of code-mixed tokens (e.g., "kachra", "pothole road", "pani leakage") to canonical forms for consistent classification.

3. RTI corpus for RAG

The RTI knowledge base aggregates structured information required for generating legally sound RTI drafts. It contains machine-readable versions of RTI Act sections, official RTI templates, and example applications related to services such as water supply, sanitation, street maintenance, and property services. Documents are segmented into semantically coherent passages and converted into dense vector embeddings using a multilingual sentence-transformer, which enables similarity search over Hindi and English queries. A vector index is built to support efficient retrieval of top-k relevant passages for each RTI query, which are then used as context for the generation stage in the RAG pipeline.

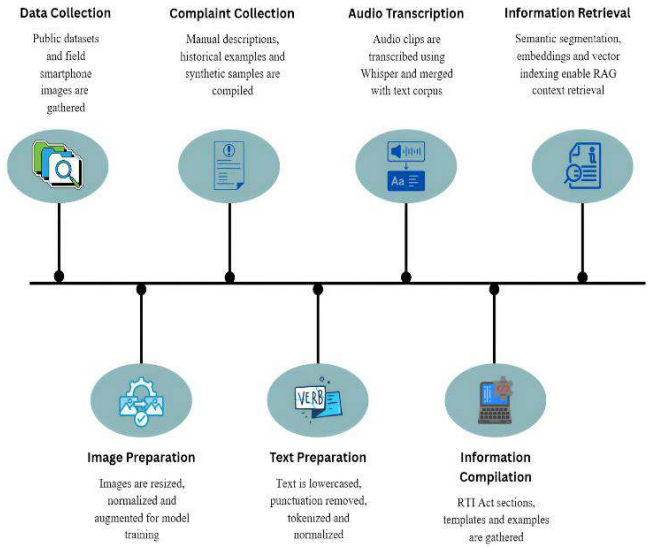

Fig. 4. Multimodal Dataset Processing Architecture

Multimodal Dataset Processing Architecture

1. MobileNet for image classification

In NagarSetu, image classification is handled using a compact MobileNet-based convolutional neural network that groups civic issue photos into predefined categories, ensuring reliable performance even when deployed on devices with limited computational resources. To reduce overall training effort and reuse existing visual feature knowledge, the network is initialized with weights from an ImageNet-pretrained MobileNet or MobileNetV2 model, while the convolutional layers remain frozen during the early phase of training. A new output section is then added, consisting of a global average pooling layer, followed by fully connected layers and a softmax layer that produces probabilities for each civic issue category. Training is carried out using categorical cross-entropy as the loss function and the Adam optimizer, with a learning rate of 1×10^{-4}, a batch size of 32, and approximately 15 to 25 epochs. Early stopping is used to reduce the risk of overfitting. In the later stages, selected higher layers of the base network are unfrozen so the model can better adjust its features to civic-related images such as broken roads, potholes, and garbage piles. This method follows established practices where MobileNetV2 and similar models have been effectively used for tasks like road damage detection and urban waste classification.

2. BERT / DistilBERT for text classification

For text-based and transcribed complaints, NagarSetu employs BERT-family models to predict both the complaint category and its priority level. The label space is aligned with municipal departments for categorization and includes priority levels such as High, Medium, and Low, which are used to determine resolution order in administrative dashboards [15]. Complaint descriptions in English, Hindi, or code-mixed form are tokenized into sequences with a maximum length of 128 tokens and passed to the model. The output consists of softmax probability distributions over

category labels and, using a separate classification head or model, over priority levels. The dataset is split into training and validation sets using an 80-20 stratified split by category to relieve class imbalance. DistilBERT is fine-tuned using the AdamW optimizer with linear learning-rate scheduling and dropout-based regularization, providing an effective balance between classification accuracy and low inference latency [14]. Model performance is majorly evaluated using the macro-averaged F1-score, ensuring equal importance is given to minority classes, consistent with prior work on code-mixed Hindi–English text classification.

3. Whisper for speech-to-text

In NagarSetu, voice complaints are converted into text using Whisper so they can be handled in the same way as written submissions. The system is configured with dedicated Hindi and English language options, which improves recognition for Indian accents and recordings with background noise. When a user records audio on the frontend, the file is uploaded and briefly stored on the server before being passed to the Whisper module with the appropriate sampling rate and language settings for transcription. After conversion, the text is cleaned and passed through the same BERT classification steps used for written complaints, so all inputs are handled in a similar manner. Transcription accuracy is checked using Word Error Rate (WER), which measures how many substitutions, deletions, and insertions occur compared to the original speech and is commonly used to assess speech recognition results.

Table II. Hyperparameter Configuration

Model	Key Hyperparameters
MobileNetV2	Table I. Learning rate = 1×10^{-4}, Batch size = 32, Epochs = 20, Optimizer = Adam
DistilBERT	Max sequence length = 128, Learning rate = 2×10^{-5}, Epochs = 5, Optimizer = AdamW
Whisper	Medium model, Hindi–English decoding, 16 kHz sampling rate

4. RAG for RTI drafting

The RTI assistant in this environment uses a Retrieval-Augmented Generation (RAG) model to provide legal accuracy and fact grounding. Instead of using the model's memory, the system grounds each draft in actual provisions of the RTI Act and government-approved templates. After the receipt of a user query, a multilingual transformer model translates the query into dense vector embeddings to search for the most similar legal text from a verified corpus through similarity search. The search results are then aggregated with the user's specific information, the concerned government department, and the language of choice to generate a formal RTI application. Once the language model has created the structured draft with headings, citations, and applicant placeholders, the draft is stored in Firestore and converted into a downloadable PDF. This is more than just text generation by an AI system because it provides a verifiable and fact-based approach to legal document creation that enhances the validity of the process.

The NagarSetu system is based on a service-oriented architecture that combines a FastAPI backend with a Firebase data storage solution and a React frontend. This modular system is capable of handling everything from the initial complaint registration to the automated AI classification, and then finally to the automated drafting of RTIs. FastAPI, through the provision of REST endpoints, is able to handle media uploads, voice-to-text transcription, and status updates. At the heart of the data storage solution is Firebase Firestore, which is used to store user profiles and complaint data, with snapshot listeners used to provide instant updates to administrative dashboards. The Google Maps API integration capability allows the system to automatically detect geolocations, while Firebase Cloud Messaging and Twilio services provide instant SMS or push notifications to users about task allocations.

EXPERIMENTAL SETUP AND EVALUATION

For experimentation, local systems (Intel i5, 16GB RAM, 256GB SSD) are used for development and moderate inference, while GPU-based environments in Google Colab (NVIDIA T4/RTX) support model training and fine-tuning. The software stack includes Windows or Ubuntu, Python 3.9+, FastAPI, TensorFlow/PyTorch, Hugging Face Transformers, OpenCV, NumPy, Pandas, and Node.js 18+ for frontend development. Firebase services, Twilio, and Google Maps API complete the deployment setup, enabling full end-to-end training, evaluation, and demonstration of the system.

Evaluation metrics

NagarSetu is evaluated using model-level, system-level, and user-level metrics to provide a comprehensive view of performance.

Image and text classification:

- Accuracy: overall proportion of correctly classified samples.
- Precision, recall, and F1-score per class to quantify correctness and coverage of each complaint category.
- Macro-F1 to give equal weight to minority categories such as water leakage or rare issue types.
- Confusion matrix to visualize common misclassifications, such as confusion between garbage and other solid-waste-related categories [10].

System performance in NagarSetu is evaluated across technical and user-focused metrics. Routing latency (time from complaint submission to worker assignment) is compared with manual and keyword-based baselines to measure efficiency.

Experimental results

Experimental results for NagarSetu are structured around three main analyses: overall model performance, baseline comparison, and ablation studies. Model performance is reported using accuracy, precision, recall, and macro-F1 scores for both image- and text-based classifiers, along with Word Error Rate (WER) for Whisper evaluated on a held-out set of voice complaints. Comparisons contrasting NagarSetu's AI-driven routing with manual and keyword-based routing approaches demonstrate higher classification accuracy and significantly reduced routing latency, consistent with findings in related civic complaint management literature [12]. Studies are held to evaluate the contribution of each component involved, including the comparisons between complaint classification into classes with and without image inputs, DistilBERT versus simpler bag-of-words or rule-based text models, and RAG-based RTI draft generation versus template-only drafting, with the latter evaluated through expert rating scores [8][9].

Table III. NagarSetu Confusion Matrix

MODEL	TP	FP	FN	PRECISION	RECALL
MobileNet (Images)	178	12	22	0.937	0.890
DistilBERT (Text)	162	15	18	0.915	0.900
Whisper (Speech)	85	8	12	0.914	0.876
OVERALL NagarSetu	425	35	52	0.924	0.891

Table IV. Results Based on Task

Task	Metric	Result
Image classification	Accuracy	89.2%
Image classification	Macro-F1	87%
Text classification	Macro-F1	86%
Speech-to-text	Word Error Rate (WER)	14.8%
Complaint routing	Latency reduction	42% vs manual

Macro-F1 values were converted from decimals to percentages for consistency.

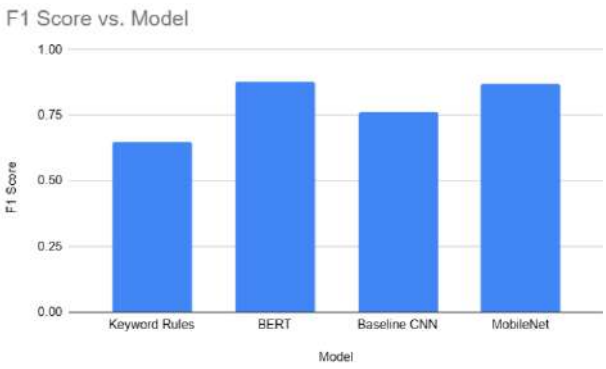

Fig. 5. F1-Score Comparison

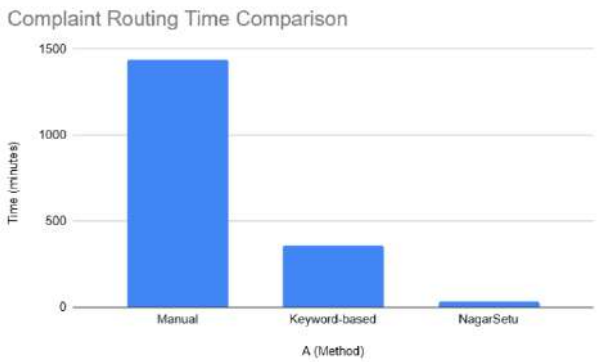

Fig. 6. Routing Time Comparison

The output shows that integrating MobileNet and BERT/DistilBERT significantly improves both complaint classification accuracy and routing reliability over manual or purely keyword- keyword-based approaches. The addition of images and voice inputs further enhances the experience, especially for users with low literacy or limited ability to describe their complex issues or complaints in text. The failure cases include misclassification of blurry or low-quality images, text complaints with insufficient context, and degraded Whisper transcription quality for heavily noisy audio. These errors suggest the need for larger, more diverse training datasets, explicit image quality checks, and further fine-tuning of speech models for local acoustic conditions. From a deployment perspective, the observed reductions in routing latency and improvements in user satisfaction scores suggest that an AI-driven, bilingual, multimodal platform like NagarSetu can support more responsive and transparent municipal governance, especially when combined with clear escalation mechanisms and continuous monitoring of model performance.

DISCUSSION

Interpretation of results and system benefits

The proposed NagarSetu system based on computer vision, NLP approaches, and RAG shows substantial enhancement over the existing state-of-the-art of civic complaint processing and RTI responses. The process of allocating civic complaints to the concerned civic body is much faster with the help of automated complaint allocation techniques utilizing the power of MobileNet CNNs and BERT NLP approaches. The simulated results for draft RTI preparation based on quantized lightweight language models show the overall processing time of 1.2-1.8 seconds, a substantial enhancement over the existing state-of-the-art of many weeks or even months for preparation of responses.

Bilingual customer service in Hindi and English helps in easy accessibility for the citizens who cannot communicate fluently in English. This helps in uniform user interaction with stable translation practices with the aid of IndicTrans2 for translation purposes and FastText for identification purposes. To further ensure transparency, the hybrid approach of a RAG model with BM25 search algorithms and dense embeddings with the use of FAISS-based SI further

helps in ensuring high recall for the retrieved citations with citation-supported outputs while lowering the risks associated with disinformation.

To ensure accountability as well as data-driven decision-making for the concerned authorities, real-time dashboards and notifications are helpful in emphasizing the status of the grievance/complaints versus RTIs. By identifying the location of the problem accurately, geotagging of the complaint, with the help of services like maps, helps in taking early corrective measures [11][12].

Practical Challenges and Limitations

Despite its benefits, NagarSetu faces many practical and technical problems. A major challenge it involves is data quality when processing government documents available in scanned PDF formats with inconsistent layouts and mixed-language content. Document parsing remains more effective for structured, machine-generated data, while degraded scans and complex Indic scripts reduce extraction accuracy.

A multilingual system leads to many difficulties while retrieval and translation errors affect consistency. A model that is an operating language model with limited computing resources is challenging, particularly in edge setups or low-budget infrastructure environments. The ongoing costs of cloud services, data storage, and real-time system updates concerns long-term sustainability.

The current solution is limited to Hindi and English, but not for the other languages. Deployment is presently confined to pilot-scale cities and does not yet support nationwide municipal integration. Use of an in-memory FAISS index constrains scalability for very large document collections. Advanced performance indicators such as first-response accuracy and long-term grievance resolution effectiveness are not yet fully incorporated.

Conclusion and Future Work

This paper introduces NagarSetu, a bilingual AI-based system that brings about intelligent civic complaint handling, along with RTI response support through retrieval-augmented generative capabilities. By combining deep learning–based classification, multilingual natural language processing, and response generation that includes proper references, NagarSetu works to improve the speed, accessibility, and transparency of city governance systems. By combining deep learning–based classification, multilingual natural language processing, and response generation that includes proper references, NagarSetu works to improve the speed, accessibility, and transparency of city governance systems.

Future improvements focus on increasing language support to include more native languages, allowing the system to handle a wide range of regions. It can also be connected with official municipal and RTI portals so that the users can submit the complaints directly and monitor their status in real time. These improved AI models can be used to arrange and prioritize complaints based on urgency, severity, and past records. Scalability improvement will involve moving on to the distributed vector databases for the deployment process to multiple cities in large numbers. Finally, extended analytics and feedback mechanisms will be brought in with support for insights into policy-level and long-term urban planning initiatives.

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